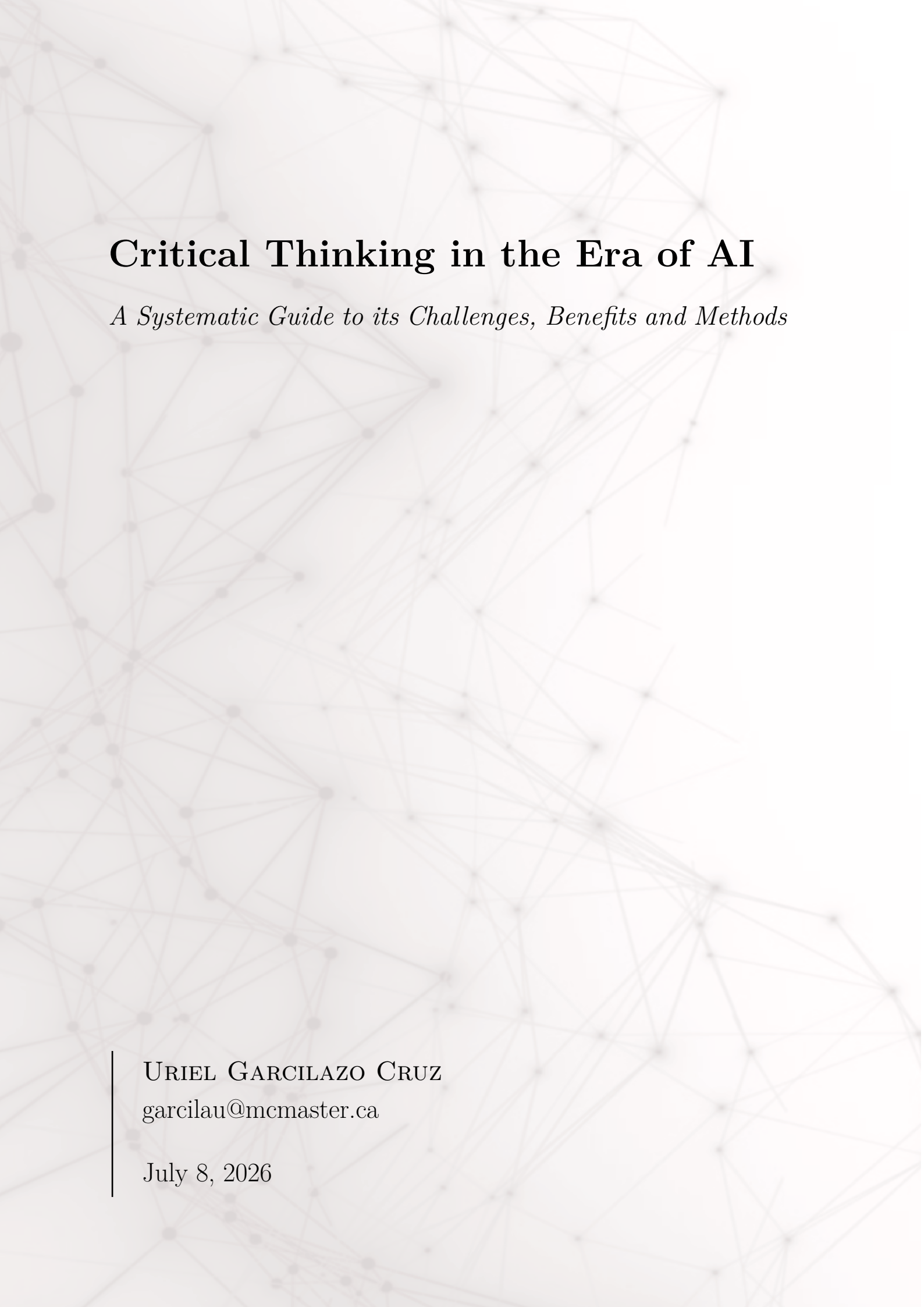


The background of the slide is a light gray network of interconnected nodes and lines, resembling a complex web or a neural network. The nodes are small circles, and the lines are thin, creating a dense, geometric pattern that fills the entire slide.

Critical Thinking in the Era of AI

A Systematic Guide to its Challenges, Benefits and Methods

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Typeset by the author with L^AT_EX.

Dedication

* * *

*In loving memory of my father, and with deepest love for my mother
and my wife*

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Introduction

My story in education began a bit more than a decade ago around 2012, when I was still developing my research at the university in Mexico City, focusing on the description of new species.. of spiders! Back then, the core of my research was the careful dissection of a spider's anatomy to identify which features could justify the existence of a species that science had not yet described. Alongside the lab work came the rest of a standard academic life: a formal education made up of countless classes, some of which were a lot less comfortable to sit through than others.

I was lucky to have the opportunity to get around incredible mentors and supervisors, many of which taught me important pedagogical tools and ways of thinking that I still carry into my work as an instructor to this day, and were particularly effective at bridging the gap between the academic life and the public narrative of scientific communication. During my years as a grad student at the University of British Columbia (UBC) I had the opportunity to embrace public outreach, where I got the chance to tell people about my research by arguing its relevance in the real world. I remember how some visitors were fascinated and others were frightened about spiders (very polarized opinions, and both valid standpoints!), and despite the polarized motivation behind the questions, the questions felt genuine and interactive. These experiences helped me understand that my position as an educator was not to judge their feelings towards my research, but to validate their emotions while simply providing evidence, hoping for a smaller gap in public's understanding as the best possible outcome.

I started spending my own time building digital media to help explain the concepts I wanted to share: media that was visually engaging and that distilled complex ideas, however simplified, into something a general audience could grasp and fall in love with (You can find some of these resources directly from my [website](#)). That's what aligned best with my professional values. Over the years I learned how to use graphic design to build board games, video games in Unity and Unreal Engine, and websites. All of it in service of effective science communication, free and out in the open, for anyone who wanted it a brief *eureka* moment of understanding in a world filled with opinionated narratives. As an experimentalist, I got fascinated by the diversity in student's attitudes towards learning, and decided to dive deeper into the rewarding, but sometimes underappreciated, field of Education Psychology, motivated also in part by my need for labeling my own struggles in Academia.

After grad school, I spent a few years refining my teaching style, working jobs

that put me in classrooms teaching programming to students of all ages. It became clear, over that time, that the methods which worked for the natural sciences like biology or chemistry (fields far less deterministic than math or code that are usually approximated via statistics) had to be adapted to the unique attributes of a new niche (computational thinking). And yet, behind the curtain of lesson planning, certain gaps in my teaching kept resurfacing regardless of the subject.

* * *

Many of my students are caught off-guard when I tell them that building a program is very similar to carefully and curiously looking at a flower or a spider. We are looking at the latest model of one of many marvelous nature innovations, and a system that has been refined over and over throughout time scales that would make the entire of human history look very humble. A different analogy is that of playing with legos. What do these things have in common?

The kind of mentality needed to write good software, or to build a robot, heavily relies on modularity and abstraction: taking small, atomic units of information, fitting them together while asking questions, and stacking them into something larger, more complex and functional (i.e. useful to solve a problem). That idea is the backbone of Software Engineering, and truly, the foundation of how we approach many problems in science.

So what is then the big fuzz with Artificial Intelligence? Why does it matter if I use it on every task? Nothing in principle! For the first time, AI allows us to process and analyze incredibly vast amounts of data, effectively allowing us to gain understanding (and influence) over increasingly higher levels of emergent behavior which undoubtedly influence the fate of the entities making the layers below. An ant colony gives a great example. A complex network of chemical signals affects every ant, and the collection of ants create a new layer of complexity that is more than the sum of its individual parts. In other words, the colony is more than the collection of ants, and in many ways it behaves as a super-organism: expanding, moving, multiplying... even the ants themselves are born with specific shapes and tasks aimed for supporting their colony! There are warriors with giant jaws, workers with long and flexible bodies, farmers, food storage balloons, and even function like doors (phragmosis)!

How does this translate into our own biology? Humans are, believe it or not, not catalogued, *sensu stricto*, as a true social species (we don't have true biological casts). Despite this we are incredibly complex social creatures with high capacity for meta cognition (thinking how to think). This reasoning capacity contributes to the assembly of a "colony" mindset where our unique interpretation of ideas and believes compounds into a functional society with unique attributes. We create leaders, infrastructure, and different ways to measure value within the group (money, real state, assets). But it turns out that our interpretation of the world is often misleading! Even when we are certain we know the truth. Our own eyes, ears and sensory signals, as biologically marvelous as they are, are also terrible tools for gathering precise data (just take off your glasses to see what I mean). And we need precision to understand the fundamental truth of the world we live in. This biased interpretation results in a variety of subjective experiences, and feeds into the believe system under which we develop, and influenced by our environment.

So what happens when a machine has the capacity for analyzing the entire data gathered by and for the colony? It quickly seems to have the capacity to provide an *unbiased* answer to every problem, every question and every sentiment that concerns the colony. Putting aside for a moment the fact that this tool is a global representation of “everything” concerning us (including our biases), even the potential of a handful of people being able to ‘steer’ opinions using such machine puts us all at risk of losing democratic control over the future of our society, simply because an element at a lower layer of complexity cannot maneuver the much larger picture that arises by emergence (i.e. our society). The machine has no trouble exploiting our craving for meaning to influence our inner dialogue, and the way social media reflects this issue is now well documented. Therefore, it is essential to be prepared at an individual level to be able to stay aligned with our own core values.

When artificial intelligence entered the picture, it arrived at a moment when our educational systems were already strained. Public schools simply did not have the capacity to serve the number of students who needed them. Personalized education felt, for most families, out of reach. Extracurricular programs offered a partial bridge, but there was no clear plan for bringing these emerging technologies into classrooms at anything like the pace they were developing in the world outside. It became obvious, in hindsight, that the educational system was lagging far behind the speed at which young students could (and did) adopt new technology, especially when it came in digital form. That lag could have a real and irreversible cost, affecting whether early-years students develop some of the most important skills there are; the skills behind critical thinking itself.

I heard this borne out in conversations with savvy parents, more than once. They would point out that their use of AI tools rests on a hybrid model (i.e. the machine doesn’t fully automate the endeavor) that supports, rather than replaces, years of education and specialization; much of it acquired at university where students are taught how to learn in the first place.

But in the era of AI, where a student can be handed nearly any answer without having to work for it, there is both opportunity and risk. On one hand, a curious, knowledge-hungry reader now has access to an extraordinary amount of information already summarized, synthesized, and tailored to their own interests. On the other hand, there is a real danger that some of the most valuable skills related to critical thinking may never take root in a generation that has grown up fully immersed in AI systems.

The wake-up call for me came when a couple of my students admitted, quite privately due to potential fear and embarrassment of their parents’ disappointment, that they were starting to wonder whether university was even worth it. This was a fair question, and to be honest, one that made me feel a bit heartbroken. Partly because of my love for the foundations of academia, but also because I felt they had a point. In the past decade alone, tuition (even for domestic students) has increased significantly without an equivalent increase in the quality of teaching — sometimes it feels the opposite — and the weight of higher education now keeps families under constant financial strain. What’s more, a student entering university is, for the first time, also becoming a functional adult: figuring out who they are while carrying the weight of a generational financial burden and time pressure unprecedented in recent generations

(it really seems we left some of the hardest challenges in our generation as parents to our own children). And now it's rather obvious that the traditional system supporting some levels of higher education (refined during the industrial era where people learned how to follow directions to be effective workers) may not be up to the task of the incoming waves of AI breakthroughs that require flexibility and real out-of-the-box thinking. By the nature of their own infrastructure universities are slow to make room for technologies students have already learned to use on their own. If anything, schools today often treat AI use as little more than a synonym for cheating, even as students are already breaking the mold and using it, mostly without guidance on how to use it well. And without that guidance, most of that incredible cognitive potential may not go anywhere productive.

And yet I still believe there is a powerful reason to move into higher education (whether it's from University or not). Not for the certification, and not for the grades, but for what a good education actually builds: a form of metacognition and meta-learning (i.e. learning how to learn), that lets a person assess the truth of incoming information by weighing the evidence behind it, often at the expense of our subjective experience. The capacity to question, objectively, is the best gift an education can give.

It became clear to me that whatever answer existed for a new kind of AI-based education, it was not going to be built inside the walls of a traditional school, or even extra curricular programs. It needed room to be flexible, and that meant changing the rules of the game. I decided to write this book hoping to assist in building a bridge for parents and young adults looking for answers on how to navigate and adapt their thinking in the era of AI. While no doubt there are many other resources available, a book still seemed like the best way to ensure a structured, systematic view of the education framework while providing statements founded on natural sciences and software engineering.

CHAPTER 1

Critical Thinking

Critical thinking is a universal cognitive skill. In science, critical thinking manifests as a specialized set of skills deeply embedded in disciplinary practice and the scientific method

What is Critical Thinking anyways?

For an instructor or parent, one of the biggest challenges resides in finding a concrete definition of Critical Thinking. Even researchers who study this for a living do not fully agree on one: some treat it primarily as a set of skills, others as a set of dispositions, intellectual habits, and others as something closer to a general way of seeing the world. We will not try to settle such debate here, but rather focus on an easier, more operational definition with the goal of providing a concrete path to help a student actually develop analytical skills.

Critical thinking (CT) can be defined, in its simplest form, as a way of reasoning where we evaluate how true a piece of information is by using logic and our understanding of the problem. We take the information apart, piece by piece, and ask whether each piece survives scrutiny when compared against what we already know. Here is an example:

“I went to the airport the other day. When we were landing I noticed there were two wheels holding the same side of the plane. What a waste! Talk about redundancy.”

Where is the fallacy? Is it that is **redundant**, or is it that is a **waste**?

As it turns out, it's neither! The person is using analytical skills to determine that having two of the same thing when you only need one is indeed redundant, and in a regular flight it is a waste, as one wheel is usually enough to take over the weight of the plane. They probably could even prove it! Screwing off and removing such annoying repetition, and landing the plane safely... at least most of the time...

The **fallacy** is in the hidden assumption that redundancy **equals** waste, as you'd obviously wouldn't put a second tire on your car. But you'd have much better chances of surviving a car accident than a plane crash. The person's claim is: “the second wheel is redundant, therefore is a waste” As it turns out, engineers are not dumb nor careless! The price of a car is nowhere near the price of an airplane, not to mention the cost to an airline's reputation if something goes wrong when landing (i.e. a tire

blows up and you blow up). The extra wheel is there to keep the plane from crashing if one of the wheels bursts. It is a planned redundancy for the sake of safety.

But the real question is this: How could our traveler know? Redundancy **can** be a waste, and in our daily lives we have gathered plenty of personal experiences that prove this — using too much toilet paper, or tapping a credit card twice for the same purchase. These experiences become habits: a default assumption about the world. So then, why should our example with an airplane be different?

Notice that even this example leans on some background knowledge: how redundancy works, why a plane is harder and more expensive to fix than a car. But it is the kind of knowledge most adults pick up incidentally, from everyday life watching the news, listening to other people stories and deriving the conclusion that it's better to be safe than sorry, a view that often pays dividends throughout our lives. That distinction matters, because the line between “general” critical thinking (some people call it common sense) and “technical” or specialized thinking is all about the information you're exposed to, and equally important, with what level of scrutiny you analyze it before incorporating it into your beliefs.

Caveats in the education system

Giving a student a dictionary and asking to learn a language is a famous example of how not to teach. In the same way, giving a student a Biology book shouldn't carry the expectation of producing a Biologist. The student needs semantics — a general and systematic framework on **how** to look at the information to make connections at increasingly emergent levels of analysis, starting from the building blocks and rooting in real, tangible outcomes to produce an enjoyable experience. I promise you this: if you want to make a biologist, take them to see and touch a whale in the wild.

In our earlier years, we depend upon our instructors to learn the basic elements and semantics of a general education, giving us a thorough (but rather simplified) view of the world. Later in college and university we learn how to autonomously query and connect new information into such a view, and in grad school this pays dividends by allowing us to identify gaps in the state of our fields, which can be solved by means of careful and rigorous experimentation.

But before I can continue glorifying education, I must issue a warning to the reader on a more realistic view of the road that may lie ahead. Education — especially higher education — is also prone to taking the path of least resistance. Many parents, students, and even collaborators already in academia often fall victims of the unspoken, yet assumed, separation of academia vs. industry, arguing towards the nobility of the knowledge-pursuing endeavor. There is some truth to that, but the uncomfortable part is this: universities are also a specialized industrial sector, with a labour structure to match a competitive market. Graduate students, who carry out much of the actual research, are exactly the kind of workforce that needs union representation. Pay scales steeply by rank: in Canada, presidents and provosts at major universities commonly earn between roughly CAD \$300K and \$600K+ annually ([Sunshine List Explorer, 2025](#)), teaching staff at research-intensive universities earn a median of roughly \$150K–\$165K ([Statistics Canada, 2024](#)), and PhD students often receive only around \$20K–\$25K per year in guaranteed minimum funding once tuition is paid — rising to \$40K annually only for the minority who win a competitive federal scholarship ([Office of](#)

the Provost, McMaster University, 2023; Liddle, 2024). Productivity metrics exist to keep researchers and their students working at full capacity (there is a reason we call the principal investigator a “supervisor”) all of it ultimately serving the university’s earnings and reputation. The result is a persistent tug-of-war between cutting costs and protecting education quality, one that, too often, leaves real damage in its wake for the students and researchers caught in the middle. I can vouch for the later. At some point during my grad studies, I had to attend a grad student course given online by an instructor playing a recording of himself explaining code. This is, arguably, the worst possible way to learn programming and felt a lot like watching a YouTube video of someone watching a recording of themselves reading a dictionary... it was torture! And definitely not worth a 3K dollar tuition. While I must clarify this is an extreme scenario, it demonstrates that education quality (at any level) is made out of a complex network of economic and cultural factors. In 2026, a student entering university is learning to navigate under an incredible amount of financial stress for the individual and their family (tuition has increased 50% for domestic students, and doubled for international students in the past 15 years (Statistics Canada, 2025, 2019)) while also entering adulthood and having to learn from an increasingly more complex world that now includes AI. We’ll dive in depth on some proposed solutions in the later chapters but it is essential for parents, educators and guardians to be aware that these pitfalls exist.

The role of STEM

Within the domains of natural sciences, technology, engineering, and mathematics (STEM), CT manifests not as an abstract intellectual exercise, but as a specialized set of cognitive moves embedded directly in the practice of the field. Put simply, this kind of CT depends heavily on the specific, technical knowledge of that field — knowledge you would not likely pick up unless you had studied the subject directly. Consider this example: knowing whether a video of a spider was generated by AI requires an understanding of actual spider anatomy — eight legs, anywhere from zero to eight eyes, fangs resting at the base of the chelicerae, and so on.

Defining CT in Science, Technology, Engineering, and Math (STEM) means shifting away from a generic “spot the flaw” framework and toward something closer to systematic problem-solving: pulling together knowledge from different technical disciplines and applying it to messy, real-world situations. Seen this way, CT in STEM is not an isolated skill a student either has or does not have — it is a form of reasoning built through genuine problem-solving, shaped by the social, cultural, and practical context in which that problem actually shows up (Nugroho et al., 2021).

There is a useful shift happening in how educators think about this kind of context. Older approaches to teaching practical skills often relied on something close to “professional realism” — essentially, having students imitate the tasks a professional would perform. Newer thinking pushes toward something richer: a framework where learning is shaped by several different kinds of context at once (Gonsalves, 2025). A general CT exercise might ask a student to spot the one logical flaw in an argument. STEM-specific CT asks for more: the student has to juggle several layers of context simultaneously — what is practically possible, what the situation calls for, what hands-on experience teaches, how different disciplines intersect, and what is ethically at stake (Gonsalves,

2025). In other words, a student reasoning critically in an engineering class is not simply plugging numbers into the right formula; they are asking how that formula holds up once it meets a messy, dynamic, real-world problem (Gonsalves, 2025). It is the difference between baking a cake from someone else’s recipe and being the one who has to invent the recipe in the first place. Both can bake the cake, but the second one is adaptable and polymorphic, capable of innovation.

In coding, mathematics, and other technical fields, this kind of thinking promises big rewards, and often takes a specific, recognizable shape known as “Computational” or “Algorithmic” Thinking. Unlike general reasoning, it offers a structured toolkit for breaking large, complex systems down into pieces a person can actually work with (Wing, 2006). The core moves in this toolkit include:

- ◊ **Decomposition:** breaking a large, complicated problem into smaller, more manageable pieces — splitting “build an app” into “design the screens,” “write the login system,” and “store the data,” for instance (Triswidrananta et al., 2022).
- ◊ **Pattern Recognition:** analyzing data to spot trends or regularities that make it predictable — noticing, say, that a website always slows down around the same time every evening (Triswidrananta et al., 2022).
- ◊ **Abstraction:** stripping away unnecessary detail to focus only on what actually matters for the solution — treating a car engine as “a box that turns fuel into motion” without worrying about every part inside it (Triswidrananta et al., 2022).
- ◊ **Algorithm Design:** creating an ordered series of logical steps to solve a specific problem — not unlike writing a recipe explicit enough that anyone, even a machine, could follow it (Triswidrananta et al., 2022).
- ◊ **Modelling and Simulation:** building a representation of a complex process so it can be visualized or tested before it is built for real — the kind of “dry run” an engineer performs before a bridge is ever constructed.
- ◊ **Evaluation:** judging whether a chosen solution actually worked, and whether that success will hold up in new, unseen situations — not just “did this fix the bug,” but “will this fix break something else down the line.”

In engineering and coding education, these six moves form the backbone of higher-order thinking. Debugging is the clearest example: a programmer hunting for an error is not merely running a technical search, but decomposing a problem and evaluating whether a fix actually holds up across different scenarios. Practicing these moves over time pushes students past rote memorization and toward something more durable — the kind of flexible, transferable thinking they will need for whatever new tool or technology comes next.

These STEM-specific frameworks do not replace the general models of critical thinking discussed elsewhere in this chapter — they sit alongside them, each suited to a different kind of problem.

In the era of AI, is being a software developer still a good idea?

Some parents and educators have raised this question with me often over the last couple of years. Advancements in AI capabilities change by the week, and for a programmer,

the idea of a machine handling every task related to writing code is an intimidating prospect. On the other hand, it offers an unprecedented opportunity: everyone can now dive into software engineering without needing to code — and I would argue that is a good thing. When a parent contacts me asking if I can teach their kid to code in a specific programming language like Python, C++, or JavaScript, I always try to steer the conversation toward programming instead: the real gateway skill that defines a software developer. What is the difference between code and programming? It's a question that takes us back to the **what** versus **how** conversation. And it is about to get weird: it turns out that a great programmer can be someone who does not even know how to code!

Here is an example of how to take a single sentence in English and turn it into a **program**:

The dog licks my spoon until there's no peanut butter left.

First, we separate the words into nouns and create definitions, simplifying complex systems into simpler versions that focus only on the elements needed to best represent the context of our statement. A dog has a nervous system, behavior, mass, volume, and so on — but the only aspect we care about in the context of our sentence is an action: licking. Therefore:

DOG is a container with one action: licking.

PEANUT_BUTTER is a number, an amount to be consumed until there is nothing left.

SPOON is trivial in this case, because there are no other objects to lick from.

Then the verbs become actions. In this case there is only one verb, which is a property of **DOG**, and which takes **PEANUT_BUTTER** and reduces its value.

Now we can rewrite the sentence in pseudo-English, replacing each word with its definition (notice the capitalization):

The `DOG.licks(PEANUT_BUTTER)` until `PEANUT_BUTTER` is 0

Finally, our program looks as follows:

```
while PEANUT_BUTTER not 0:
    DOG.licks(PEANUT_BUTTER)
```

The pseudocode above describes the **algorithm**: the sequence of logical steps that encode the essence of our task. Notice that we didn't need special knowledge on a programming language to derive it. Instead, we systematically applied a form of *Decomposition* and *Pattern Recognition* by categorizing and stripping words from english to derive the core elements needed. We applied *Abstraction* by vastly simplifying what a dog and peanut butter is, and used these new information motifs to *Model* our algorithm.

Our algorithm can be coded into any programming language of our choice. The algorithm itself does not change — only its syntax does. Below is that exact same algorithm, written out in three different languages. Notice that the **what** (a dog, an

amount of peanut butter, and a licking action that repeats until the peanut butter is gone) stays identical in all three; only the **how** (the specific words and punctuation each language requires) is different.

Python

```
class Dog:
    def licks(self, peanut_butter_amount):
        peanut_butter_amount -= 1
        return peanut_butter_amount

dog = Dog()
peanut_butter_amount = 5

while peanut_butter_amount != 0:
    dog.licks(peanut_butter_amount)
```

JavaScript

```
class Dog {
    licks(peanut_butter_amount) {
        return peanut_butter_amount - 1;
    }
}

const dog = new Dog();
peanut_butter_amount = 5;

while (peanut_butter_amount !== 0) {
    peanut_butter_amount = dog.licks(peanut_butter_amount);
}
```

C++

```
#include <iostream>
#include <cstdint>

class Dog {
public:
    void licks(int64_t& peanut_butter_amount) {
        peanut_butter_amount -= 1;
    }
};

int main() {
    Dog dog;
    int64_t peanut_butter_amount = 5;
    while (peanut_butter_amount != 0) {
        dog.licks(peanut_butter_amount);
    }
    return 0;
}
```

Three languages, three sets of rules, three syntaxes to memorize — but one algorithm. This is precisely the distinction between coding and programming: coding is learning the syntax and rules of a specific language; programming is a separate process of designing the recipe in the first place, and is the real reason why we call people that write algorithms Software Developers. Remember, a programming language is for a programmer what a calculator is to an accountant; just a tool.

An LLM (agent or chatbot) excels at following instructions and writing the coding portion, and they have begun to demonstrate a very capable level of programmatic reasoning as well, capable of setting goals, evaluating correctness and even making assumptions over missing information from the dossier. For a traditional Software Developer (probably a person in their 40's or above) this is problematic, because we are used to have a tight grip over the architecture of the code — i.e. meticulously inspecting the syntax to ensure the correct implementation of our algorithms, which in turn provides a clear understanding on the limitations of such program: what type of inputs break the program, what are the boundaries of the data being fed, and how to properly build a modular integration that can be plugged into a bigger ecosystem. The idea of losing control is somewhat overwhelming.

A story about change

Does the development of AI agents mean the end of Software Developers? The answer to the question is perhaps a murky yes and no. Experiential data from older colleagues in the areas of Biology, Chemistry and Computer Science reveal that the profile of any career is dynamic, with a tendency to move from a narrow niche into increasingly broader areas, and sometimes merging with other fields. Perhaps the reason why AI feels so overwhelming is caused by the fact that a Biologist can now write programs for Chemistry, a Physicist can dive into Biology, and everybody will be expected to be a Software Developer. Where do we even start? Whenever there's uncertainty about the future, it's always wise to look at the past, and the field of Taxonomy is a great case for me because it was experiential, living it myself from the 2000's to the present, being told to me by amazing researchers who lived theirs from the 80's and 90's, who themselves heard and told stories from the "golden" expedition era from the beginning of the 20th century.

Back then, the word Biologist wasn't even a thing. A person with a keen eye and a passion for describing and cataloging the biodiversity of animals and plants was called a Naturalist. If you really liked spiders, beetles, ants, or any other cryptic organism, you would spend most of your time going to exotic places (if you or your friends had money) to find them in their natural habitats and describe them: how do they look? What do they eat? What do we name its kind?

Every animal species we know of today has followed roughly the same protocol since: describe it, and give it a name following the rules of zoological nomenclature. Those rules are anchored to a single starting date, 1 January 1758, treated as the publication date of the 10th edition of Linnaeus's *Systema Naturae* — and, by a deliberate quirk of the rules made specifically for my own corner of biology, also the publication date of Carl Clerck's *Aranei Svecici*, the founding work of spider taxonomy, even though it most likely actually appeared in 1757 ([International Commission on Zoological Nomenclature, 1999a](#)). The official rulebook itself — what we now call the International Code of Zoological Nomenclature — came much later: the first international rules were only adopted in 1905, with the modern Code established in 1961 ([International Commission on Zoological Nomenclature, 1999b](#)). Plants and fungi, for what it's worth, play by a different rulebook entirely, with their own separate starting date of 1753. Some species have since disappeared, and others have been renamed, but this methodical approach to species description was called **Taxonomy**. A taxonomist was a specialist in a group of organisms.

Then, in the early 1900's, a new field began brewing. Scientists had already consolidated powerful statistical methods capable of outlining patterns in the data that could be used as arguments over the possible relationships among species. It would take several more decades, though, before they had the machines to put those methods to work at scale: the first electronic computers did not exist until the mid-1940's, and it was really the 1950's and 1960's that brought computers into universities, finally letting scientists process amounts of data they could not have analyzed by hand. The profile of a Taxonomist changed. It was no longer just describing species, but also being capable of gathering evidence to feed to the machines, and being able to analyze the results under scrutiny and scientific rigor. The field of Taxonomy + Comparative

Biology transitioned into **Systematics**: the scientific study of biological diversity and the evolutionary relationships among organisms over time.

We learned, among other things, that “fish” are not a real, valid group in the cladistic sense (yeah, you heard that right!). Because a fish gave rise to tetrapods (amphibians, reptiles, birds, and mammals, including us), making a group that includes fishes without us is like drawing a family tree but excluding your weird uncle Larry. For scientific purposes, Larry is still a Johnson and is very much worth studying.

The Systematics Wars

Once statistics and computers entered the picture, systematists did not simply agree on how to use them. Instead, the field fractured into competing schools of thought, each with its own answer to a deceptively simple question: what makes a classification *correct*? The conflict that followed, fought across journals, conferences, entire academic careers and a lot of *Yo Mama* jokes from the 1960’s through the 1980’s, is remembered in the history of science literature as the **Systematics Wars** (Stern and Lidgard, 2017).

On one side stood **phenetics** (also called numerical taxonomy), founded by Robert Sokal and Peter Sneath in their 1963 book *Principles of Numerical Taxonomy* (Sokal and Sneath, 1963). Pheneticists argued that classification should be based on overall measurable similarity — as many characters as possible, measured and fed into a computer, which would then cluster organisms by how alike they were, with no assumptions about evolutionary history required. This is the **all-in approach** on computation reliance.

On the other side stood **cladistics**, built on the work of German entomologist Willi Hennig, whose 1950 book was translated into English only in 1966 as *Phylogenetic Systematics* (Hennig, 1966). Hennig insisted that classification should instead be based strictly on *shared derived characters* — features that indicate genuine common ancestry — and that the only acceptable groupings were monophyletic ones, exactly the standard that ruled “fish” out as a valid category above, but was completely valid for pheneticists.

A third camp, **evolutionary systematics**, represented by established figures like Ernst Mayr, tried to hold a middle ground: classifications should reflect evolutionary history, but could also weigh the overall degree of difference between groups, even at the cost of occasionally allowing controversial groups.

This was not merely a polite academic disagreement. It was, by most contemporary and historical accounts, genuinely personal: careers were made and blocked, symposia turned hostile, and the rivalry was infamous enough that the historian and philosopher of science David Hull devoted an entire book, *Science as a Process* (1988), to chronicling it as a case study in how scientific communities actually behave under pressure (Stern and Lidgard, 2017). By the 1990’s and 2000’s, cladistics — increasingly folded into the broader label of **phylogenetic systematics** — had become the dominant paradigm in the field, helped along enormously by a new kind of evidence that was about to make the debate almost irrelevant: DNA.

The Role of DNA

Up to this point, every character used to classify an organism — a leg, a fang, a vein pattern on a wing — came from its anatomy. Starting in the late 1970's and through the 1980's, systematists gained access to an entirely new kind of character: the genetic code itself.

Two technical breakthroughs made this possible. The first was DNA sequencing: in 1977, Frederick Sanger and colleagues published the chain-termination method, the technique that made it practical to actually read out a DNA sequence base by base (Sanger et al., 1977). The second was the polymerase chain reaction (PCR), first demonstrated in 1985 by Randall Saiki, Kary Mullis, and colleagues, which solved a much more basic problem: how do you get *enough* DNA to sequence in the first place, starting from a single tiny tissue sample (Saiki et al., 1985)? Together, sequencing and amplification meant a systematist no longer needed a live colony or a freezer full of tissue — a leg from a single pinned specimen could, in principle, be enough.

This changed the character of the Systematics Wars more than any argument could. DNA sequences gave cladists exactly the kind of objective, quantifiable, heritable characters Hennig's method demanded, but at a scale morphology could never match: thousands of base positions per gene, instead of a handful of visible traits. Molecular data routinely confirmed relationships that decades of morphological argument had failed to settle, and just as often overturned them outright — it is molecular evidence, as much as cladistic theory, that nailed down the status of “fish” beyond reasonable dispute. New statistical machinery arrived alongside the data to make sense of it: maximum likelihood methods for estimating trees directly from DNA sequences, formalized by Joseph Felsenstein in 1981 (Felsenstein, 1981), and Bayesian approaches that followed in the years after. **Being a Systematist now meant something close to being a molecular biologist and a statistician**, on top of everything a Taxonomist already had to be.

The Transition to Genomic-Scale Data

For roughly two decades, molecular systematics still meant Sanger sequencing: slow, expensive enough that a single study might responsibly analyze one gene, or a handful, across all of its species. That changed again in the mid-2000's with the arrival of next-generation sequencing (NGS) technologies, which replaced one-sequence-at-a-time Sanger reactions with the ability to sequence millions of DNA fragments simultaneously, at a small fraction of the cost per base (Young and Gillung, 2020).

The effect on systematics was, once again, a complete change in what the job required. A single study could now plausibly include hundreds or thousands of genes, entire transcriptomes, or whole genomes, across dozens of species at once — an approach now called **phylogenomics**. But genome-scale data did not just mean *more* of the same kind of analysis; it introduced computational problems that simply did not exist at the scale of one or a few genes: aligning thousands of sequences instead of one, identifying which genes in different species are actually the same gene (orthology), handling enormous amounts of missing or low-quality data, and running analyses that increasingly require dedicated computing clusters rather than a single researcher's laptop (Young and Gillung, 2020).

Let's put all in perspective. A modern Systematist, then, is not just someone who can recognize a spider by its eye pattern, or run a statistics package, or read and extract a DNA sequence — each of which used to be an entire career back in the day. They are also, by necessity, something close to a bioinformatician and a software developer — writing pipelines to process genomic data nobody could have imagined handling by hand a generation ago. The taxonomist became a systematist; the systematist became, in no small part, a programmer. In a way that is easy to miss for people outside the field, the profile of a person just wanting to study an animal gradually became more inclusive in the set of skills needed to answer larger and more consequential questions that affected all of Biology.

Software Development in the era of AI

Our previous section illustrates how the profile of a career can change over time, and it is an important example because it demonstrates how technology forces multidisciplinary integration into an otherwise narrow and specialized field. Software developers feel this at an accelerated pace which receives much more attention from the media because it sits at the heart of the industry, putting it at the front row of innovation. And it is beginning to show a new type of debate: how much of the old to give away to adapt to the new? New and especially young software enthusiasts understandably advertise and quickly adopt new terms like *vibe coding* and *prompt engineering*, filling hours of content in platforms like YouTube with new ways to use emergent technology. In contrast, older programmers and developers are more reluctant to let go of the old ways, adapting at a slower pace and also gradually adopting the AI wave of innovation, although from a more skeptical standpoint, full with traceability and verifiability, compromising speed for a clear, well-structured code that is modular and scalable (i.e. not monolithic), putting the *science* in Computer Science.

My take is that the process is in a maturing state that eventually will settle into a hybrid model that tolerates rapid change. We remember that critical thinking in algorithmic endeavors relies in *abstraction*, *modularity* and *pattern recognition*, and for the first time a process that seemed slow and technical can be worked around using agents, starting from the installation of packages compatible with hardware, or very low abstraction forms of programming like Assembly, GPU optimization or parallelization, to name a few. If you are a scientist looking to explore classic algorithms like ant colony optimization, dynamic programming or wave function collapse, with an application in mind, then is possible to delegate the responsibility of the algorithm's integration to an agent (although with some guardrails and ensuring traceability). The programming side shines in this new environment while the coding portion is taken care of by an agent specialized in code that ensures proper syntax.

From Rule-Followers to Goal-Seekers

The earliest AI “agents” used in programming were nothing like the ones making headlines today. They worked off a fixed rulebook: a human wrote out every condition the agent might encounter and exactly what to do in each case, in frameworks researchers called Belief-Desire-Intention systems. Picture a very elaborate flowchart — the agent could follow it perfectly, but only within the boundaries someone else had already drawn. It could not improvise.

The 2000s and 2010s brought a different kind of agent: one that learned from trial and error instead of from a rulebook. Programs like TD-Gammon learned to play backgammon by playing itself millions of times and gradually favoring whatever moves led to a win. This is the same instinct a child develops after losing enough games of tic-tac-toe. Multi-agent systems built on frameworks like JADE let several of these learners coordinate with each other, directing traffic lights or industrial robots. Around the same time, voice assistants like Siri and Alexa matured into something genuinely useful for everyday conversation. But every one of these systems, however capable, was still bounded: each had learned to do one job well, and none of them could step back and reason about a brand-new problem the way a person can.

The 2020s changed that. Large language model agents — with names like Auto-GPT, Devin, and Codex-based coding assistants — can take a high-level goal stated in plain English, break it down into steps no one wrote out in advance, and then go execute those steps: cloning a code repository, editing files across the project, running the tests, fixing what breaks, and committing the result, all without a human reviewing every line along the way (Sapkota et al., 2025). This is the leap from a system that follows instructions to one that sets its own sub-goals on the way to a larger one; from a rule-follower to something closer to a goal-seeker.

The Speed of the Shift and the One-person army

What makes this transition feel so abrupt is how fast it moved. In 2021, the open question in the field was whether a language model could reliably autocomplete a single function. By 2026, frontier coding agents are resolving roughly four out of five real bug reports and feature requests pulled directly from mature, real-world code repositories — the kind of work that used to define an entire engineer’s day (Bhati, 2026). The center of gravity has shifted from *code generation*, the simple act of producing text that looks like working code, to what researchers now call *delegated execution under human supervision*: the agent plans the work, carries it out, checks its own output, and reports back, while a person decides what to delegate in the first place and reviews what comes back (Bhati, 2026).

This has already produced a new kind of job. Instead of one engineer writing one piece of software, a single specialist can now direct several AI agents at once, with some planning the work, others carrying it out. It is the equivalent of how a construction foreman might direct several specialized crews working different rooms of a house simultaneously, checking blueprints and signing off on finished work rather than laying every brick personally. Using one such method, a single engineer directing somewhere between fifty and a hundred coordinated agents has been shown to complete 75 to 150 tasks in a single day. Needless to say, this output would previously have required a team of eight to ten human engineers (Carbowitz and Kumar, 2026). Whatever you call this role — orchestrator, director, conductor — it did not exist five years ago, and it is not quite “software developer” in the sense anyone would have meant in 2015.

The Tug-of-War

As with every transition in this chapter, rapid adoption has not settled the argument about whether this is, on balance, a good thing. There is a real tug-of-war here between people racing to delegate as much as possible to these agents, and people warning that something important gets lost along the way.

One concern is what researchers call the *technical-debt hypothesis*: because asking an AI agent to write more code costs almost nothing, agents tend to patch problems locally rather than redesign anything properly, the same way it is always tempting to wrap a leaking pipe in tape instead of re-plumbing the wall behind it. Tape is faster today and more expensive tomorrow, and a codebase patched this way over and over can quietly degrade even as it keeps shipping features (Bhati, 2026).

A second, subtler concern is what happens to the person doing the delegating. When an AI assistant reliably produces good-looking output, it is easy to mistake

that output for your own growing skill — the same way relying on GPS turn-by-turn directions for years can make you very good at following a voice telling you where to turn, without ever actually learning to navigate the city yourself. The day the signal drops, the gap in your own ability is suddenly very real. Researchers describe something similar happening with AI-assisted work: the quality of what comes out stops reliably tracking the competence of the person who asked for it, which means it becomes harder, not easier, to tell who has actually built real skill and who has only built a habit of asking well (Koch, 2026b).

Put those two effects together and a two-track outcome starts to emerge. Engineers who already have deep experience tend to use these agents the way a master carpenter uses a power tool: it does not replace their judgment, it multiplies it, freeing them to focus on the architecture and the decisions that actually require expertise. Engineers who are still early in their careers face the opposite risk: using the agent to *finish* the work rather than to *learn* the work, which can look identical in the short term and very different five years later (Bhati, 2026).

All of this reveals a pattern many professionals encounter at some point in their careers: the title on someone’s job description rarely keeps pace with what they actually spend their day doing. The taxonomist became a systematist, then something close to a bioinformatician. The programmer became a software engineer, then split into web developers, data engineers, and DevOps engineers. What today’s developer is becoming — orchestrator, director, the person who decides what *not* to delegate — is simply the next name in that same long line of trades that defines the new profile of a “Software Developer”.

What Vibe Coding Actually Is

We have spent real time making the case that a great programmer does not necessarily need to know how to code — that programming, the act of designing the algorithm, is what matters, and coding is just one possible “how.” A new style of working with AI, nicknamed **vibe coding**, takes that idea and pushes it about as far as it can go: the developer steers entirely through plain-English description and intuition, in something closer to a back-and-forth conversation with the machine than a structured handoff of formal requirements (Sapkota et al., 2025).

Think of the difference between cooking a meal yourself from a recipe, and describing the meal you want to a chef and trusting them to figure out the rest. A vibe coder does the second. They are not writing syntax, debugging a compiler error, or thinking about how memory gets allocated; they are describing, in natural language, roughly what they want, and then judging the result by whether it “feels” right — whether the app looks correct, behaves the way they expected, does the thing they asked for — rather than *building the code together* (Sapkota et al., 2025).

This shifts the actual skill being tested. Where it used to take years to master a language’s particular quirks, the bottleneck now is how well someone can phrase a request. And the accuracy of those initial prompts turn out to be critical: these AI tools are highly sensitive to exactly how a problem is worded, and a small change in phrasing or a missing piece of information in the semantics of the algorithm can produce a noticeably different piece of code (Pudari and Ernst, 2023). The danger is to release a program into the world that wasn’t properly tested, and could behave

unexpectedly under circumstances not foreseen by the prompter, potentially putting the users at risk. And yet, despite all of the warnings, it truly is so much fun to play with! The appeal of working this way is obvious once you see it in action. Without the usual step of sketching out an architecture before writing a line of code, a vibe coder can build the interface and the underlying logic at the same time, shaping both as they go, and giving a great deal of freedom and experimentation, which are critical aspects of STEM education. It is closer to the way a sculptor might work with clay rather than follow a blueprint (Spinner et al., 2026). The loop becomes: state an intent, look at what the AI produced, decide if it “feels” acceptable, and either keep it or ask again. A working prototype that once took real planning and setup to assemble can now take a single sitting.

There is no such thing as a free lunch

All of that speed is real, but I’d warn the reader to ensure “vibe coding” is treated as a learning platform, and not as an ideology for writing *real* software (i.e. destined to be used to solve real problems). I’d argue that vibe coding is a prequel that leads to a more matured form of prompt-based engineering (i.e. one that involves Algorithmic Thinking, as we defined it at the beginning of this chapter) later, used the same way an accountant uses a calculator. Vibe coding could show benefits in the development of critical thinking (CT), but only when results can be systematically and pedagogically analyzed. Not doing so carries dangers to the student and their surroundings.

The first cost is structural. We already saw that AI coding agents are biased toward adding more code rather than redesigning anything, because writing new code is cheap and a proper redesign is expensive. Vibe coding, with no formal architecture in place to push back against that bias, tends to make it worse: the codebase can quietly bloat with every new request (Bhati, 2026). Worse, without any deliberate structure holding the pieces apart, a vibe-coded project can turn into what researchers call an opaque, tightly coupled monolith — imagine inheriting a machine where every wire is soldered directly to every other wire, with no diagram anywhere showing which connects to what. The machine works, right up until something breaks, and then nobody, not even the person who built it, can tell which connection to check first (Spinner et al., 2026). Even if the program can be saved, the cost of rewiring everything is more expensive than the cost of building it from scratch.

The second cost is security, and it is a sharper one. A vibe coder who never learned to read code closely has no real way to catch a problem hiding inside something that otherwise looks fine. Current AI tools are still maturing to the point where they can reliably flag when a piece of generated code does something risky, like quietly deleting or exposing sensitive files, which means software can end up shipped on the strength of its “vibe” — it seems to work — with nobody ever checking whether it is actually safe (Priyadharasini et al., 2024).

The third cost is the one this chapter keeps circling back to: what happens to the person doing the delegating. Vibe coding does not just change how experienced engineers work; it lowers the door enough that people can start building software without ever picking up the fundamentals at all. That is precisely the trap researchers describe as automation bias — trusting an output simply because it appeared, without the habit of checking it (Pudari and Ernst, 2023). And because we skip the part where

the struggle actually teaches you something, we can end up able to produce a working prototype while still lacking the deeper skills that those same systems cannot do for them (Bhati, 2026).

How to Use AI to Improve Critical Thinking

If AI is now woven into how people learn, how do you make sure it builds critical thinking rather than quietly replacing it? This is an active area of research, and a handful of concrete strategies keep showing up.

Make the AI's Mistakes Visible

The biggest risk in handing a student AI-generated code is that they treat the result the way you might treat a vending machine: press a button, take the snack, never think about what happened inside. One classroom system that combines block-based coding with a robot and a conversational AI tutor tackled this head-on by making the robot's behavior fully inspectable: every time a student's program ran, the system displayed a step-by-step execution trace, so students could see exactly where the robot's actual behavior diverged from what they had planned, instead of just being told "it didn't work" (Li et al., 2026). Students stopped treating bugs as mysterious failures and started treating them as a trail to follow back to its source.

A related strategy goes a step further: instead of waiting for AI mistakes to happen naturally, some researchers deliberately plant them. In a pilot university programming curriculum, instructors occasionally inject a small, carefully chosen error into an AI-assisted assignment — not to trick students, but to give them practice diagnosing and repairing exactly the kind of mistake a real AI tool will eventually hand them on its own (Dranias and Whitley, 2026). It is the same logic as a driving instructor who quietly anticipates a hazard on a quiet road instead of grabbing the wheel the instant something looks off: the goal is a student who notices the problem themselves, not one who has simply never been allowed to make one.

Build Courses That Outlast the Tool

We argued earlier in this book that a great programmer is really someone who has mastered the algorithm, not the syntax — that the "how" of any particular language is the part that changes, while the "what" underneath survives the move from one tool to the next. The same principle turns out to matter for how AI gets taught in schools. A curriculum built around the quirks of this year's chatbot is obsolete the moment a better chatbot ships; researchers studying AI-assisted programming education have argued that grounding instruction in durable concepts — system design and control abstractions — rather than tool-specific tricks is what lets a curriculum survive rapid technological change (Dranias and Whitley, 2026). In practice, that means leaning on the core components of computational thinking that show up again and again in the research: decomposition, pattern recognition, abstraction, and the rest (Ihsan et al., 2022).

This also means leaning on teaching methods that were never about a specific tool to begin with. Project-based learning and design thinking — having students work through a real problem from start to finish, using AI to speed up implementation rather than replace understanding — give students the full arc of building something, not just the AI-generated middle of it (Rodrigues et al., 2024). It also means resisting the urge to silo AI inside computer science class: when teachers from history, English, and

engineering co-design AI lessons together, students get to apply their own discipline’s way of thinking to AI-generated material, which exposes both what the AI got right and where it quietly went wrong (Xie et al., 2024a). And it means giving teachers real room to rearrange their own syllabi — research on critical-thinking curricula more broadly has found that the teachers who get permission to deviate from a rigid, prescribed structure are the ones who manage to give critical thinking equal billing with whatever technical skill is nominally being taught (Le and Hockey, 2021).

Plan Before You Prompt

One of the clearest findings from researchers studying AI-assisted work, discussed earlier in this chapter, is the “crutch effect”: when an AI tool reliably produces good-looking output, people start mistaking that output for their own growing skill, and short-term performance stops predicting what they can actually do when the tool is no longer available (Koch, 2026b). The most direct countermeasure educators have proposed is almost embarrassingly simple: make students commit to their own plan *before* they are allowed to ask the AI for anything.

One pedagogical framework built around this idea breaks the work into six phases: **plan**, **prompt**, **preview**, **produce**, **peer-review**, and **portfolio-tracking**. The crucial discipline lives in the first two steps — students sketch their own approach to the problem before they ever open the AI tool, and then compare what the AI actually produced against that original plan, rather than the other way around (Kong et al., 2024). That ordering is what keeps the thinking the student’s own: it is the difference between checking your work against an answer key after genuinely attempting the problem, and copying the answer key first and reverse-engineering a justification.

Make the Test AI Can’t Fake

All of the above falls apart if assessment does not change to match it. A test that only grades the final block of code is, by now, a test of how good a student is at prompting, not how well they understand the problem — and that gap is exactly what AI exposes (Gonsalves, 2025). The fix researchers point to is to grade the layers AI struggles to fake: **process**, **justification**, and the **reasoning trail** that led to the answer, not just whether the answer is correct against a key, or whether it compiles and runs (Singh et al., 2026).

None of this works without the people running the classroom. Surveyed students consistently say they do not want AI to replace their teachers; they want a mentor-led partnership, with explicit help learning to spot an AI’s hallucinations and blind spots rather than just being handed the tool and told to figure it out (Singh et al., 2026). That puts the real bottleneck on teacher preparation: professional development that walks teachers through the practical, classroom-level work of building and grading AI-aware assignments, not just a one-off seminar on what a chatbot is (Forsström and Bond, 2024), paired with the same kind of collaborative, theory-grounded training that has long underpinned good critical-thinking instruction in general (Franco and Vieira, 2019).

Keeping Rigor in AI-Driven STEM Prototyping

Everything in this chapter so far has been about software, but the same forces are re-shaping science and engineering labs. A physicist running fluid-dynamics simulations, a biologist modeling drug responses, a structural engineer optimizing a bridge design — all of them are increasingly working alongside AI agents that can write and run code on their behalf. The stakes are different here than in ordinary software, though. A bug in a to-do-list app is an inconvenience; a bug in a structural simulation or a drug-interaction model can harm people. STEM practice needs additional considerations in the classroom and in the field.

AI as Orchestrator, Not Just Generator

One of the most useful pattern researchers have found is to use AI as an **agent**. An agent can be considered a specialized chatbot that remembers what it's doing and is able to call tools with the intention of expanding its understanding of a problem and to solve said problem. We can think about it as a master chef calling out to the grill station or the pastry station rather than personally cooking every dish. In one fluid-dynamics study, researchers wrapped an entire physics simulator behind an interface an AI agent could query directly — instead of running a full, expensive simulation for every parameter it wanted to test, the agent could ask the wrapped simulator for a near-instant estimate, probe thousands of configurations in the time a single traditional simulation would have taken, and use what it learned to decide where to dig deeper (Vishwasrao et al., 2026). The AI is not generating physics from nothing; it is orchestrating a fast, narrow tool (the simulator) and reasoning about the results.

A related pattern shows up in biology: rather than asking an AI to predict every outcome in one shot, which tends to produce overconfident, unreliable guesses, some systems work step by step, the way a detective solves the easy parts of a case first and uses that solid ground to tackle the harder parts. One gene-perturbation system predicts the easiest, highest-confidence outcomes first, then uses those confirmed results to inform its guesses about the harder, more ambiguous ones, rather than guessing at everything simultaneously (Kim et al., 2026). A conversational drug-discovery tool applies the same logic to the question itself, breaking one complicated scientific query into several narrower, more answerable ones before running anything (Süwer et al., 2026).

None of this works without boundaries. The same kind of structure we described earlier in this chapter — explicit contracts spelling out what each component is responsible for, agreed before any work gets delegated — shows up again here, letting researchers keep control over the decisions that matter (what question to ask, what counts as an acceptable answer) while handing the AI the repetitive parts (Spinner et al., 2026).

Show Your Work

A simulation that looks plausible is not the same thing as a simulation that is correct, and the difference matters enormously more in STEM than it does in a typical app. The fix researchers keep landing on is essentially the same one your high school

chemistry teacher insisted on: showing your work. One hardware-and-software engineering framework requires that any AI-generated result come with what amounts to a lab notebook: simulation logs, the specific tool versions used, timing records, and formal checks, so that the result can actually be audited and replicated rather than just trusted (Koch, 2026a). For especially high-stakes work, the same framework keeps the AI agent that builds something separate from the AI agent that checks it, on the theory that an agent grading its own work is exactly the same conflict of interest that arises from a student grading their own work (Koch, 2026a).

A broader survey of this kind of “harness engineering” makes a related point: the AI’s own claim that something worked is not evidence that it worked. A properly built system has to independently verify that the world actually changed the way the AI said it would — checking the simulator’s actual internal state, not just accepting the AI’s summary of what it did — and when that verification fails, use the mismatch itself as the diagnostic signal for what went wrong (Ning et al., 2026). Some of this validation work can itself be automated: AI testing agents can run validation checks continuously, including while no human is watching, freeing researchers to focus on the higher-level question of whether the experiment is asking the right thing in the first place (Feldt et al., 2023). None of this removes the need for a human in the loop, though — one study of generative AI in agile engineering workflows found it could usefully reprioritize which checks happen first as requirements shifted, but cautioned that the underlying suggestions still needed a specialist to weigh them against the actual variability of the project and its team, not blind trust (Bahi et al., 2024).

Where It Breaks

Two failure modes show up again and again in this research, and both should be familiar.

The first is the crutch effect we discussed with software developers, and it applies just as directly to scientists (after all, we’re students too!). When an AI consistently produces confident-sounding output, researchers can lose the calibration signal that would normally tell them how good their own independent judgment actually is — the professional STEM version of mistaking a fluent answer for a correct one, simply because nothing about how it was delivered hinted otherwise (Koch, 2026b). In a STEM context, this is the difference between a simulation that runs without errors and a simulation that is physically possible; an AI agent has no built-in instinct for which one it has produced, and neither, eventually, does a researcher who has stopped checking.

The second is the technical-debt problem, also already familiar: because generating more code is cheap and properly redesigning a flawed approach is expensive, AI agents default to patching rather than rethinking, and the same researchers who first identified this pattern in ordinary software engineering note that if an agent can turn out roughly ten plausible-looking patches an hour, the actual bottleneck quietly shifts to whoever has to review them (Bhati, 2026). In scientific computing specifically, this same bias toward quick patches can manifest as code that runs cleanly but encodes something the discipline itself would call nonsense. For example, a query that returns an answer with no biological plausibility, for instance, rather than one that simply fails to run (Süwer et al., 2026).

What This Looks Like in Practice

The growing sophistication of digital tools used in STEM research points to the importance of a clear theoretical foundation, one that can guide both the handling of data and the decisions behind a system’s software architecture. Where the boundary between success and failure is clearly defined, and the underlying data is already standardized or preprocessed, that clarity significantly boosts the efficiency of the agentic tools working with it. A few concrete examples make the pattern easier to see. In circuit design, a new simulation tool built explicitly with AI sizing agents in mind — the kind that need to test thousands of component configurations very quickly — achieves large speedups by carefully avoiding a bottleneck where multiple processors try to update the same piece of data at once, but its authors are candid that the speedup also depends on parts of the system nobody had gotten around to optimizing yet (Bao and Chitnis, 2026). In materials science, a topology-optimization tool for programmable gel structures leans on a neural network to predict material behavior, paired with a mathematical shortcut that avoids re-deriving expensive calculations from scratch every time a parameter changes (Chandrasekhar et al., 2026). And in computational chemistry, a neural model built to speed up standard quantum-chemistry calculations works well on the molecules it was trained on, but its own authors note it gets noticeably less reliable on larger, more complex molecules — exactly the caveat a careful researcher needs to see stated plainly, rather than discovering it the hard way (Wu et al., 2026).

What ties all three together is not the specific science, but the discipline around it: speed without accountability, and the lack of an honest account of where the tool’s confidence starts to run out and begins to *assume* results in a tool that is not actually useful at all! And running these types of protocols in laboratories, where resources are often expensive, only leads to frustration. Again, it seems like human intervention is essential for the proper calibration of experimental design.

CHAPTER 2

The Stages of Thinking

The way we think changes with age. Middle Childhood children require concrete examples and repetition on tangible experiences. Adolescence sees the main transition into formal operational thinking, characterized by higher levels of abstraction and capabilities in metacognitive analysis that leads to self regulation and self assessment

Our previous chapter traced how AI is restructuring professional work by providing historical perspective on how trades and professions change with technology, how it lands in specific areas of STEM like software development, and gave an introduction on how new AI technologies can and are being used in real labs. This chapter is about what is happening inside the person sitting at the desk: the child who types a homework question into a chatbot and takes the answer as given, the teenager who lets an AI plan their essay and then is surprised when they struggle to talk about what it says. The question this chapter asks is not what AI is doing to jobs. It is what AI is doing to the development of the thinking that makes jobs possible in the first place.

The Building Blocks of Critical Thinking

Researchers who study critical thinking tend to describe it not as a single ability but as a family of *dispositions*: habits of mind that distinguish careful, reflective reasoning from the kind that accepts the first plausible-sounding thing it encounters (Usanova et al., 2024). Critical thinking requires evaluating whether a claim is actually supported by evidence, recognizing when a line of argument is circular or manipulative, and putting pieces of information together in a way that produces something genuinely useful rather than just something that sounds right. What makes it difficult to teach is that it resists the kinds of activities that dominate most classroom time. It cannot be drilled. It requires encountering real uncertainty and working through it, which is exactly the kind of friction that a tool good at producing confident, fluent answers tends to remove.

Metacognition sits directly underneath critical thinking. It is the ability to think about your own thinking — to step back from a task and ask whether the approach is working, whether the conclusion actually makes sense, whether something important has been missed. Researchers usually break it into three phases: planning before starting, monitoring comprehension and progress while working, and evaluating whether the outcome was any good afterward (Mitasari and Hidayah, 2022). What is consistently striking in the research is how much these capacities resist passive development. They

do not consolidate on their own; they require contexts that force the learner to make their reasoning explicit — to notice that something has gone wrong and work out where (Suarningtyas, 2022). Problem-based learning environments, which place students in contact with a real problem before handing them the tools to solve it, are among the settings that most reliably build metacognitive skills, precisely because they refuse to resolve the difficulty prematurely.

Self-regulated learning is the layer above both: the ability to set your own goals, choose appropriate strategies, and adjust when something is not working, without being managed by an outside agent. This is the developmental peak of the three-capacity system, and it is where the relationship with AI introduces its most uncomfortable complications. A recent study of students using an AI-powered study system found a striking pattern: the AI's capacity to generate ready-made study paths and pre-sorted materials reduced what the researchers called the *preparatory burden*: the effortful, time-consuming setup that students used to do themselves. The problem is that this burden, frustrating as it was, was also doing something important. Students who no longer had to organize their own material reported losing what the researchers called *cognitive ownership*: the sense that what they were studying was genuinely theirs. The study found that editing and modifying AI-generated content was not merely a correction step — it was the mechanism by which information became personal knowledge. Removing it left students more efficient on paper and less capable underneath (sma, 2026).

Computational thinking — decomposing a problem into parts, recognizing patterns, abstracting away irrelevant detail, designing a step-by-step approach, and evaluating whether a solution actually holds up — offers a concrete window into all three of these capacities in operation (Ihsan et al., 2022). What research on computational thinking instruction keeps finding is that these components are not independent items to tick off; they cluster together (Khenner, 2024). A student who struggles with algorithmic thinking tends to struggle with decomposition and pattern recognition as well, suggesting they are facets of the same developing cognitive architecture rather than separable skills.

The practical implication for any adult thinking about a child's development is subtle but important. What children and adolescents actually need from learning environments is not to have their problems solved for them, but to have difficulty made navigable while they are still the ones doing the solving. The educational concept behind this is old and well-established: the most productive instruction happens in the space between what a learner can do alone and what they can do with temporary support, and the support has to thin out over time for genuine capacity to build (Tiantong and Teemuangsai, 2013). An AI tool that permanently removes cognitive load rather than reducing it while the learner grows is not a scaffold. It is more like a crutch, which makes the work of walking easier, but also weakens the leg the longer it is used.

Middle Childhood: Learning by Doing First

Children between roughly six and eleven years old are in the middle of one of the most important transitions in human cognitive development. They can follow rules, sort and classify, understand cause and effect, and persist through multi-step tasks. What they largely cannot yet do is reason fluently about things they cannot see, touch, or encounter directly. Their thinking remains anchored to concrete, tangible experience, and that anchorage is not a limitation to overcome as quickly as possible; it is the foundation that makes later abstract thought possible.

The effects of this in classrooms is well-documented. Elementary mathematics structure learning in stages, which builds motivation and connects to what students already know, core practice that moves from teacher-guided to independent work, and a closing that requires reflection on what was just done (Sutama et al., 2024). What researchers studying these environments keep finding is that skipping the concrete stages — giving children unfamiliar problems and asking them to work independently before the underlying concept has been built — tends to produce anxiety rather than learning, and often leads students to simply not engage. There is reward in repetition.

The emergence of early scientific thinking follows a similar pattern. Young children who are given structured opportunities to notice things, test simple hypotheses, and compare outcomes are already developing the seeds of the kind of evaluative reasoning that critical thinking depends on. They cannot yet identify logical fallacies in an abstract argument, but they can observe that their prediction was wrong and wonder why. Research on K-12 scientific thinking instruction suggests that this earlier, more experiential groundwork is exactly what later analytical skills plug into: when secondary students struggle to think critically about arguments, it is often because the earlier scaffolding was never built (Schmaltz et al., 2017). Teaching scientific thinking, meaning the practice of asking “how do we actually know this?” and “what would change my mind?”, begins earlier than most curricula acknowledge. STEM integration in early childhood settings shows that children are capable of genuine inquiry-based learning from quite young ages when the environment is built to support it (Intisari et al., 2024).

Metacognition begins to emerge distinctly during this period, though in a form that is almost entirely externalized. A six-year-old (usually) does not spontaneously step back from a math problem and evaluate whether their approach is working; they need a teacher to prompt that kind of reflection explicitly. What the research on metacognitive development in elementary settings suggests is that structured problem-based tasks — where students have to articulate their plan before starting and then revisit it after — can meaningfully accelerate that internalization process (Mitasari and Hidayah, 2022). The goal is not just better scores, but rather building the habit of checking your own work against your own intent, which is a capacity that turns out to matter enormously in every context of the adulthood.

The complication that AI introduces at this stage is not subtle. Children are naturally inclined to reach for resolution — to find out what the answer is and move on. An AI that provides instant, confident-sounding answers to almost any question a curious eight-year-old can think of is, from a developmental perspective, a very efficient engine for skipping exactly the part that builds the capacity, making supervision critical

from parents and guardians. The research on self-regulated learning makes clear that what consolidates self-regulation during middle childhood is precisely the experience of managing the discomfort of not knowing, making a plan anyway, and finding out whether it worked (Suarningtyas, 2022). Tools that eliminate that discomfort — rather than helping children navigate it — are not helping children grow. They are removing the conditions growth requires.

The Adolescent Mind: When Abstraction Takes Hold

The shift that happens between roughly twelve and eighteen is not primarily about getting smarter. It is about gaining access to a mode of thinking that was genuinely unavailable before. Adolescents who are progressing normally through this period begin to be able to reason about hypothetical situations — things that are not in front of them and may never be. They can hold multiple competing propositions in mind at once, evaluate which is better supported based on their own interests and values, and even consider what it would take to change their view. This is the stage Piaget called *formal operational thinking*, and its emergence is the prerequisite for the kind of critical reasoning that higher education and professional life both demand.

The transition is not a switch that flips. It is gradual, uneven, and heavily dependent on the contexts in which it is practiced. Adolescents who are regularly asked to work through ill-structured problems, the kind that cannot be solved by applying a memorized rule, develop formal operational capacity faster than those who are not, because it is the encounter with genuine complexity that drives the development (Suarningtyas, 2022). One study of computational thinking instruction using robotics found that students shifted from treating errors as failures to treating them as diagnostic signals: rather than restarting when something went wrong, they traced the execution step by step to isolate exactly where their algorithm and the robot's behavior had diverged (Li et al., 2026). That shift from frustration to diagnosis is a good functional description of what formal operational thinking actually enables.

Interdisciplinary approaches that ask students to apply analytical tools across subjects tend to accelerate these gains, because they force the reuse of abstract patterns in new domains. A student who practices pattern recognition in mathematics and then encounters the same underlying logical structure in a history argument about causation has done something cognitively different than a student who learned those patterns in only one place (Utary and Anwar, 2023). Reviews of computational thinking instruction across higher education contexts suggest that the skills cluster together precisely because they are expressions of the same underlying capacity (Khenner, 2024): learning to decompose a problem in one subject makes decomposition more available in every subject.

Self-regulated learning deepens significantly during adolescence, moving from the relatively external, prompted form it takes in elementary school toward something genuinely internal. The three-phase structure — planning and goal-setting before starting, maintaining focus and managing effort during, and reflecting and adjusting afterward — becomes more autonomous as the teenager matures (Yuniarti and Astuti, 2022). What intervenes in this development is the balance between challenge and

support. When the support system consumes too much cognitive bandwidth (i.e. when a student is managing the scaffolding rather than managing the task) it can produce anxiety that actually impairs the self-regulatory process it was meant to support (sma, 2026). The most productive support system during this period is simple, transparent and modifiable: students understand what it is doing and can adjust or remove it as their own capacity grows.

Metacognitive depth expands considerably during these years. An early adolescent is becoming able to monitor not just whether a task is done but whether the approach to it was well-chosen, and to articulate the difference between knowing how to execute a procedure and understanding why it works. Research on metacognitive interventions in secondary contexts consistently finds that structured activities requiring students to evaluate their own understanding — comparing alternative solution paths, explaining their reasoning aloud, revisiting a conclusion in light of new information — produce significant gains in exactly these capacities (Ehis and Blessing, 2020). Metacognitive monitoring is foundational to self-regulated learning: it is the mechanism by which a student detects that something is not working before the problem compounds (Mitasaki and Hidayah, 2022). The adolescent who has developed this monitoring capacity does not just perform better on assessments. They are better equipped to learn anything new because they have a functioning internal indicator telling them when their understanding is solid and when it only feels that way.

What AI Does to All of This

Chapter 1 described a pattern that shows up repeatedly in the research on AI-assisted professional work: the output gets better while the underlying skill quietly atrophies. The same dynamic is at work in learning environments, and it is worth understanding precisely how.

When an AI tool generates a confident, fluent answer, it does not just resolve the question the student asked. It also creates the impression of understanding where there may be none. Researchers studying AI-mediated self-assessment have described this as “metacognitive decoupling”: the quality of what a student produces becomes disconnected from what they actually know, because the AI’s fluency transfers to the student’s confidence without the student having done the cognitive work that confidence is supposed to reflect (Koch, 2026b). The consequence is what the same research calls a “crutch effect”: a student who has calibrated their confidence against AI-assisted output does not merely lack skills without the AI. They have also lost the calibration signal that would normally tell them how dependent on the tool they have become. Short-term performance metrics stop predicting what a student can actually do independently, which is precisely the thing education is supposed to develop (Koch, 2026b).

This does not mean AI is simply bad for learning. Thoughtfully designed AI can function as a genuine support tool that makes difficulty manageable without eliminating it. AI-driven debugging hints that show students where their code’s behavior diverged from their intent, without handing over the corrected version, keep the diagnostic work with the student (Li et al., 2026). Pedagogical frameworks that require

students to plan before they prompt, and to compare the AI's output against their prior plan rather than accepting it wholesale, preserve the cognitive ownership that development requires (Kong et al., 2024). The question is not whether to use AI in learning contexts. It is whether the specific design of the interaction preserves or bypasses the work that builds capacity.

The research on AI literacy in student populations is revealing about what this looks like from the student's side. One large-scale study of students interacting naturally with a generative AI system found that they spontaneously developed five distinct patterns of use. Not just "asking questions" but navigating breakdowns, editing outputs, and negotiating with the system about what counted as an acceptable response (cha, 2026). The students who developed the most sophisticated AI literacy were not the ones who used the AI most fluently. They were the ones who treated the AI's outputs as material to work with rather than conclusions to accept, and who understood themselves as the final judge of accuracy and relevance, not the AI. This is what researchers mean when they distinguish "usage familiarity" from "critical familiarity": the first means knowing how to operate a tool, and the second means understanding what the tool does and does not know how to do (Singh et al., 2026).

AI also interacts with the institutional structure of learning in ways that complicate individual development. When curricula are built around fixed learning objectives assessed through standardized outputs, AI can produce outputs that meet those criteria without requiring the reasoning they were designed to assess. This is not a problem with any particular tool; it is a structural problem created by the mismatch between what AI can generate and what assessments have historically measured. Research on assessment design in the AI era argues that the solution is not more restrictive supervision but more contextually complex tasks: assessments that embed personal experience, real-world dilemmas, and disciplinary nuance in ways that require the student's own reasoning to navigate, not just a well-prompted AI's (Gonsalves, 2025).

What this means practically is that designing AI well for learning environments — whether at home or in school — requires keeping two things in view simultaneously: what the AI is doing for the learner right now, and what it is doing to the learner's developing capacity over time. These are not always the same thing. An AI that produces a perfect summary of a chapter is helping a student in the moment and potentially harming them over a year, if summaries are what was supposed to develop their ability to read closely and synthesize argument.

The research on critical reflection with AI tools suggests that the most important design feature is transparency: tools that expose their reasoning rather than just their outputs, that acknowledge uncertainty rather than projecting uniform confidence, that invite the user to question the result rather than accept it (Glinka and Müller-Birn, 2023). Students who learn to interact with AI this way, exactly as a collaborator whose suggestions require evaluation rather than an oracle whose outputs are final, are developing exactly the critical disposition needed to be autonomous thinkers (cop, 2026).

What This Means for Parents and Educators

The preceding four sections contain a fair amount of developmental theory. Here is the practical argument that follows from it.

The core risk is not that AI will somehow make children completely incapable of critical thought. Rather, AI excels at removing the very experiences that make understanding feel personal, meaningful, and rewarding. After all, what incentive would a brilliant detective like Sherlock Holmes have if the killer had simply left their name and address on a note at the crime scene? The effort of planning before starting. The discomfort of not knowing. The diagnostic work of figuring out where a plan went wrong. The metacognitive checking of whether what you think you understand is actually solid. All of these can be quietly outsourced to AI tools that are designed to be helpful.

The implication is not to ban AI from children's lives, which is neither realistic nor necessarily desirable. AI tools used well can provide the kind of targeted, responsive scaffolding that a single teacher in a classroom of thirty students cannot always provide. They can make feedback fast enough to be pedagogically useful. They can give a student another explanation when the first one did not land. The implication is to be deliberate about which parts of the work remain with the student. Not as a matter of principle but as a philosophy of teaching, one that cleverly makes the teacher's life easier by providing a hybrid model where the agent is a Teaching Assistant.

Translating this into practice perhaps looks as follows: an engaging introduction is followed by a plan that then becomes the prompt, and the AI's output is compared against the student's own prior thinking and expectations (Kong et al., 2024). Was the result unexpected? What assumptions were introduced by AI? Where are those assumptions coming from, and are they correct? The resulting conversation can be fed back into a loop that integrates students guided by a teacher, and using AI as an experiment that helps reveal patterns of a phenomenon or mechanism under study. This loop rewards guided assessment, not just conclusions, and attaches credit to the process of getting there, not just the finished product (Gonsalves, 2025). However, it is essential for teachers to be supported with real curriculum development time and real co-design of AI-integrated lessons, not just a list of approved tools handed down from a district office (Van Brummelen and Lin, 2020), and developing curricula designed around durable cognitive objectives rather than any particular tool, so that the core learning remains intact when this year's chatbot is replaced by the next one (Dranias and Whitley, 2026).

None of this resolves the harder question of how schools and families make these decisions under genuine time and resource constraints, when the AI tool is free and the metacognitive intervention costs something. That is the territory the next chapters will explore. But the foundation the argument rests on is what this chapter has tried to establish: critical thinking, metacognition, and self-regulated learning are not decorative skills that education produces on top of scores. They are the real work, which develop through specific and well documented kinds of experiences. And the tools that are currently entering every home and classroom can either help pushing education forward or stagnate it.

CHAPTER 3

AI Suitability based on Age

The previous chapter argued that critical thinking, metacognition, and self-regulation develop through specific kinds of experience. We also argued that AI tools, when not used deliberately, can quietly replace those experiences with something that looks similar but produces different developmental results. This chapter sharpens that argument by asking who, exactly, is at risk and why. The answer is not uniform across age groups. A teenager who has already developed some capacity for metacognitive monitoring and abstract reasoning faces a different set of risks than a seven-year-old who has not. An adult who understands, at least roughly, that AI systems are probabilistic text generators faces a different set of risks than a child who has no framework for what she is talking to. Understanding these differences is not just theoretically interesting; it is the precondition for deciding how to act.

The Early Years

The primary school years (roughly ages five to twelve) are a period of rapid construction. Children are not simply learning facts during this time; they are building the cognitive architecture that will determine how they handle new information for the rest of their lives. Research on coding instruction in kindergarten settings finds that even very young children can make significant gains in computational thinking skills when given structured curricula, which suggests the cognitive architecture for logical reasoning is malleable and present earlier than often assumed (Bers et al., 2023). The implication is not that AI should be kept away from young children entirely, but that what is done with it during these years matters disproportionately, because these are the years when foundational structures are either built or not built.

Mathematics instruction in elementary school offers a useful model for thinking about this. Effective elementary math teaching follows a careful progression: opening activities that connect new material to what students already know, core exercises that move from demonstrated to guided to independent practice, and closing activities that require the student to reflect on what was just learned (Sutama et al., 2024). What teachers who have worked in these environments know — and what the research confirms — is that skipping the independent practice phase, where the student confronts the problem alone and has to figure something out, tends to produce students who appear to have learned the material but cannot actually apply it when the external support is removed.

What makes the primary school years particularly important is that children at

this age are generally not yet capable of compensating for what they miss. Older students and adults who encounter a tool that is doing too much for them can recognize this, step back, and add the missing work back in. Primary school children, whose executive functions are still maturing and whose metacognitive capacities are still largely externalized and dependent upon authority figures do not have the internal resources to make that correction (Suarningtyas, 2022). The tool fills the gap, the gap that was supposed to build the capacity, and the capacity does not develop.

When the Tool Does the Thinking

The specific mechanism by which AI tends to undermine this development goes by several names in the research literature, but the core mechanism is simple. When a child encounters a difficult problem, the traditional learning process requires effort: retrieving what they know, trying an approach, noticing it does not work, revising. This effort is not inefficiency, but rather how information gets encoded in long-term memory. The struggle that feels unproductive is doing the pedagogical work. One would be surprised on how often managers and workers (responsible adults) in the industry forget this simple lesson, prioritizing volume and losing control and perspective over their own products.

As we mentioned in previous chapters, generative AI is disruptive because it provides immediate, fluent and confident-sounding resolution. The child who uses AI without proper supervision no longer has to try an approach and fail if the primary concern is the solution (i.e. the grade). The result is what researchers have called a *fluency illusion*: the ease with which the student can read the AI's response is mistaken, by the student and sometimes by an outside observer, for actual understanding (sma, 2026). The student has not encoded anything because there was nothing to encode through; the AI did the cognitive work that encoding requires.

This produces what research on AI-assisted learning has called the crutch effect. When an AI tool reliably generates good-quality outputs, the person using it begins to calibrate their confidence against the output quality rather than against their own developing competence. *Self-assessment starts to track the AI, not the person's independent ability* (Koch, 2026b). For adults, this is a professional risk. For primary school children, it is a developmental one: the calibration signals that normally tell a learner "I have mastered this" or "I am still missing something" are being replaced by signals that tell them "the AI produced something that looked right." The two things are not the same, and conflating them during the years when foundational cognitive structures are being built means those structures may simply not form.

The SmartFlash study of AI-assisted studying found something instructive about the relationship between AI assistance and cognitive ownership. When students were given AI-generated content they could not edit or modify, they disengaged from the material in a way they did not when the content was their own work or when they had meaningfully modified an AI draft (sma, 2026). Editing turned out to be doing something more than correction; it was the process by which external information became internalized knowledge. For a child in the primary school years who is still learning how to construct an argument, organize a piece of writing, or work through a mathe-

mathematical problem, being handed a finished version denies them the central experience. They get the output without the process that makes the output meaningful.

Research on generative AI in classroom settings identifies a related concern: that AI literacy alone — knowing how to prompt a tool, how to evaluate whether its output is plausible — is necessary but not sufficient. What students also need is metacognitive training: the capacity to detect a gap in their own understanding, to notice when they do not actually know something despite being surrounded by text that describes it (Abdelghani et al., 2023). Curiosity-driven learning depends on this capacity, because curiosity is essentially a response to the awareness of a knowledge gap. A tool that eliminates the awareness of the gap eliminates the curiosity that the gap was supposed to generate.

The Trust Problem

There is a second mechanism by which AI poses a distinct developmental risk to younger children, and it operates through the social cognition that children bring to all their interactions with apparent agents. Humans are fundamentally social learners: we rely on other people as sources of information, and we calibrate how much to trust those people based on cues about their reliability and benevolence (Hoehl et al., 2024). Children are, if anything, more sophisticated about this calibration than adults in some respects — they track whether an informant has been right before, they are sensitive to cues about an agent’s competence and helpfulness — but they also apply this framework in ways that are developmentally specific.

One of those specificities is that children, particularly in early and middle childhood, have a marked tendency to over-attribute social qualities to objects that exhibit human-like communicative behavior (Hoehl et al., 2024). A generative AI system that responds in natural language, addresses the child by name, and produces confident, friendly outputs is going to activate the child’s social cognition in a way that a calculator does not. The result, documented across multiple studies, is that children are more likely to accept information from an AI agent not because they have evaluated its accuracy, but because the interaction pattern resembles a trustworthy informant. The “halo” of conversational fluency functions, for a child, the way the halo of apparent confidence and authority functions in human social learning.

Adults also exhibit trust in AI that exceeds what the evidence warrants, but through a different mechanism. For adults, the risk is what researchers call *epistemic over-trust*: accepting the content of AI outputs because they appear authoritative and internally coherent, even in domains where the person lacks the expertise to verify them (Hoehl et al., 2024). For children, the risk is anterior to content: they may not even reach the point of evaluating accuracy because the interaction itself has been read as socially trustworthy. These are different problems requiring different solutions.

The developmental research on children’s trust in apparent authorities is relevant here. Children as young as five show the same kinds of appearance-based trust biases that characterize adult social judgment, suggesting these biases emerge relatively early and are robust (Ewing et al., 2019). An AI interface designed with friendly, confident, human-like features will activate these trust dispositions automatically and in a way

that precedes any evaluation of the AI's accuracy. This is not a criticism of any particular design choice; it is a description of how young children process social information. The implication for anyone deploying AI tools with young children is that the interaction design is itself a pedagogical decision, regardless of whether it is recognized as one.

What students who develop genuine AI literacy look like is different from students who are merely fluent users. Research on naturalistic AI interaction found that sophisticated AI literacy manifests as students treating AI outputs as material to work with and evaluate, rather than conclusions to accept: they negotiate with the system, ask follow-up questions, catch inconsistencies, and treat AI-generated text as a starting point rather than an endpoint (cha, 2026). This is exactly the disposition that children's social trust tendencies work against, because social trust operates by simplifying the evaluation process, not by making it more demanding. Developing genuine critical familiarity with AI requires teaching children to apply something like the opposite of their default social response.

Not Just Another Screen

It is worth being specific about how the risks described above differ from the concerns about screen time and digital media that have been part of educational and parenting discussions for decades. That distinction matters because the interventions appropriate for general screen time risks — limits on duration, attention to whether content is passive or active, encouragement of face-to-face interaction — are necessary but not sufficient for the AI-specific risks.

General screen time research has found that the quality and interactivity of the content matters considerably. Unstructured, passive consumption is associated with attention difficulties; active, structured digital engagement can support learning. Research on STEM in early childhood education finds that the same device can have very different effects depending on whether it is used for passive entertainment or for inquiry-based exploration (Intisari et al., 2024). Blended learning approaches that integrate digital tools with face-to-face instruction and reflective activities can build critical thinking skills when designed well (Chetry, 2024).

What makes AI different is not primarily the medium or the duration of exposure. It is the nature of what the AI does during the interaction: it generates, it synthesizes, it resolves. Traditional passive screen time gives children content that already existed and was made by other people for other purposes. AI systems, at the request of the child, produce content that appears personalized, purposeful, and directly responsive to the child's expressed need. The child is not watching something that someone else made; the AI is making something for them, in real time, in response to their specific question. This makes it more seductive and, for purposes of cognitive development, potentially more damaging — because the cognitive work that the child was supposed to do has now been done, in the form that makes it look as though the child did it.

The risk of cognitive offloading is therefore more pronounced with AI than with other digital media. A child watching an educational video about how volcanoes work is consuming content that was made before they arrived; they still have to make sense

of it, remember it, connect it to what they already know. A child who asks an AI to explain how volcanoes work and receives a four-paragraph, clear, accurate, appropriately pitched response has been given something that mimics the product of their own comprehension without requiring them to go through the process that comprehension involves. The monitoring capability in computational thinking environments confirms that when AI feedback is available but physical or concrete engagement is absent, younger learners tend to accept outputs without genuinely working through the reasoning (Li et al., 2026).

Possible Concrete Frameworks to Support Implementation

None of the above is an argument for keeping AI out of children's lives, any more than the concerns about passive television watching were arguments for keeping televisions out of homes. The argument is for deliberateness: for understanding what AI is doing and what it is not doing, and designing the interaction accordingly.

At the level of the home, the most effective form of adult involvement is active rather than restrictive. Research on children's interaction with digital technology consistently finds that the **quality of parental engagement** matters more than the presence or absence of limits. For AI specifically, what active mediation looks like is helping children develop the habit of checking the AI's outputs rather than accepting them: asking "do you think that's right?", "how could you find out if it's wrong?", "what would you say if you were explaining this yourself?" (Van Brummelen and Lin, 2020). These questions push the metacognitive work back to the child. They require the child to engage with the AI's output as material to evaluate rather than a conclusion to copy.

Privacy is also a genuine concern at this age, distinct from the cognitive risks but equally real. Children's interactions with AI systems are frequently logged and may feed back into the system's training. A developing framework for LLM applications designed specifically for children proposes privacy-by-design principles that include strict data minimization, session-based memory isolation (so the system does not build a persistent model of the child), and guardian-accessible oversight of what information is being collected (pri, 2026). These are design principles that parents should be asking about when evaluating AI tools for their children, and that schools should be requiring when selecting platforms.

At the level of classroom design, the most important shift is in how assessment is structured. An assessment that only grades the finished product — the essay, the solved problem, the completed project — is measuring a combination of the child's capability and whatever AI contributed, not the child's capability alone. Assessment that requires children to explain their reasoning, defend their choices, or demonstrate understanding in a novel context is measuring something much closer to actual independent competence (Gonsalves, 2025). This requires more design effort from teachers, but it is the kind of effort that makes assessment meaningful rather than just efficient.

For younger learners particularly, the most effective AI-integrated activities tend to be those that preserve the "productive struggle" that learning requires. Computa-

tional thinking curricula that use AI tools to provide targeted hints — showing students where their logic diverged from expected behavior, rather than simply producing the correct answer — keep the diagnostic work with the student (Li et al., 2026). Pedagogical frameworks that require students to plan their approach before interacting with AI, and then compare the AI’s output to their prior plan, preserve the cognitive sequence that planning ability requires (Kong et al., 2024). Design principles for AI learning tools that are grounded in educational research consistently emphasize that effective scaffolding must transfer responsibility gradually, not eliminate it (Luckin and Cukurova, 2019).

For adolescents, who have more developed metacognitive capacities and can engage with AI in more sophisticated ways, the relevant skills shift toward what researchers call orchestration: the ability to decompose a complex task, judge which parts can be usefully delegated to AI, evaluate AI outputs critically, and retain judgment over the whole (Bhati, 2026). These are learnable skills, but they require explicit instruction — they do not emerge automatically from use. A teenager who uses AI tools for three years without being taught to think about what she is delegating and what she is retaining will not necessarily develop orchestration skills; she will become a fluent user without the critical distance that genuine competence requires (Singh et al., 2026).

The equity dimension of all of this cannot be left unmentioned. Access to AI tools that are well-designed, safe, and genuinely educational is not evenly distributed. When better-resourced students have access to AI tools that scaffold and challenge rather than just produce, and less-resourced students have access only to tools that produce, the gap between them widens even if both groups appear, from the outside, to be “using AI.” The teacher co-design research is clear that effective AI integration requires educators to be involved not just as implementers of tools designed by others, but as participants in the design of how those tools are used (Van Brummelen and Lin, 2020). The professional development investment this requires is not a luxury item; it is part of what determines whether AI favors equality or undermines it.

The practical principles that follow from this chapter are modest in their ambition and demanding in their execution: **Keep the cognitive work with the child wherever possible.** Ask what is being outsourced and whether that was work the child needed to do. Design assessments that require the child’s reasoning, not just the child’s output. Choose tools whose design is transparent and whose privacy practices are appropriate for the age group. And recognize that the medium has changed but the developmental needs have not: children still learn by doing, still need to struggle, still need to check their own understanding and find it wanting and try again. A tool that makes all of that unnecessary is not, however convenient, a learning tool.

Metacognition and Self-Regulation in Young Learners

One of the better-established findings in educational psychology is that the ability to accurately judge what you know is itself a skill that develops over time, and that it develops slowly, unevenly, and in ways that are systematically vulnerable to the wrong kinds of feedback.

Adults typically show a mild overconfidence bias: they think they understand things

slightly better than they do, and their judgment of their own learning correlates reasonably well with how they actually perform on tests. Children, particularly in the primary and early secondary years, show something more extreme. Their self-assessments often bear very little relationship to their actual competence. This is not because children are dishonest or careless; it is because accurate self-assessment requires a schema for evaluation that has not yet been fully built. To know whether you understand something well, you need a model of what genuine understanding looks like — which requires enough prior learning experience to have calibrated that model. Young learners lack this calibration, and so their confidence tends to track something else: how fluent the material feels, how familiar it seems, how easily they can read an explanation and nod along.

This is the metacognitive mechanism behind what researchers sometimes call the Dunning-Kruger dynamic in young learners. The pattern in children is somewhat different from the adult version: it is not primarily that incompetent students overestimate themselves because they cannot recognize incompetence. It is that all students, competent and incompetent alike, tend to miscalibrate because they are using the wrong signal. Fluency of processing is mistaken for depth of understanding. Research on multimedia instruction makes this concrete: participants who learned from rich video materials rated their understanding significantly higher than those who learned from static images, even though their actual test performance was essentially identical (mul, 2024). The vividness of the experience was doing something cognitively — it felt like learning — but it was not producing better retention or transfer.

AI-generated answers are, in this sense, extremely fluent. They tend to be well-organized, clearly expressed, appropriate in register, and correct in the most salient details. A student reading an AI explanation of photosynthesis, or an AI-written paragraph about the causes of the First World War, is very likely to feel like they have learned something — because the output is fluent in exactly the way that triggers the “I understand this” signal. What the output is less likely to do is force the student to retrieve prior knowledge, encounter their own gaps, or work out why something is true rather than just that it is. The research on metacognitive training in secondary school settings finds that structured Problem Based Learning (PBL) interventions can significantly improve the accuracy of students’ self-assessments in planning, monitoring, and evaluating, but only when those interventions explicitly require students to predict their outcomes and then compare their predictions against actual results (Mitasari and Hidayah, 2022). The accuracy comes from the mismatch, not from the answer. An AI that reliably produces well-matched answers removes the mismatch, and with it, the calibration signal.

This matters especially for scientific and analytical domains where overconfidence has downstream consequences. Students who do not realize they have misunderstood a concept do not seek clarification; they proceed on the basis of a flawed model and often cannot diagnose the resulting errors. Research on the development of scientific thinking skills finds that the first step in teaching genuine scientific reasoning is teaching students to notice their own cognitive biases and the places where their intuitions conflict with evidence (Schmaltz et al., 2017). Students who are consistently handed fluent outputs (regardless of the medium) never have to develop the habit of noticing the conflict. The gap between what they feel they know and what they actually know

simply does not become visible.

Productive Struggle and the Paradox of Difficulty

There is a well-established set of findings in learning science that goes by the name of *desirable difficulties*: learning conditions that temporarily make performance worse but ultimately produce better retention and transfer. Spacing study sessions over time rather than massing them together. Interleaving different topics rather than practicing one thing until it feels mastered. Requiring learners to generate an answer before seeing the correct version. All of these make learning feel harder and slower, and all of them produce substantially better long-term outcomes than the alternatives that feel easier and faster.

The reason these difficulties work is that they force the kind of effortful cognitive processing that produces robust memory traces. When a student has to retrieve something from memory — even if they retrieve it incorrectly and then see the right answer — the retrieval attempt itself strengthens the eventual memory of the correct information. When a student has to switch between topics that require different mental frameworks, the act of switching and reorienting consolidates each framework. The cognitive struggle is doing the work.

The developmental challenge is that younger learners have limited capacity to recognize this dynamic. A primary school student who is finding material difficult is likely to conclude that the material is too hard or that they are not good at it, rather than associate the difficulty as a sign that learning is happening. This is not a fixed feature of younger minds; it is a consequence of having limited experience with the relationship between effort and learning. Scaffolding (the temporary support that helps students manage difficulty without eliminating it) is the established pedagogical mechanism for bridging this gap, with the critical requirement that the support must fade as the learner develops capacity (Tiantong and Teemuangsai, 2013). The goal of scaffolding is not to make things easy; it is to make difficulty survivable while the underlying capacity is being built.

In this context, AI tools used for homework or problem-solving without deliberate design have a tendency to function as permanent, non-fading scaffolds. They do not thin out over time; they are as available on the day before an exam as on the first day a student encounters a concept. Research on academic self-regulation and procrastination finds that secondary students with low strategic self-regulation (i.e. those who have high motivation but poor metacognitive planning) are significantly more prone to avoidance and procrastination, even when they genuinely want to succeed (Valenzuela et al., 2020). The executive function deficits that AI tools reduce in the short term are, in many cases, exactly what these students need to develop.

The neurodiverse population is worth noting specifically here. Students with attention or executive function differences often experience cognitive load that is genuinely higher than their peers, and they benefit from scaffolding that is more persistent and more carefully designed. But the principle that scaffolding should be timed to the learner's developing capacity rather than permanently applies equally. The main difference is in the scaffolding parameters and not in the underlying logic (Azuka et al., 2024). Inclusive instructional design for neurodiverse learners emphasizes reducing anxiety and creating conditions where students feel safe to struggle instead of elimi-

nating struggle as such.

Teaching the Skill of Learning Oneself

The evidence on metacognitive interventions converges on a straightforward finding: metacognition can be taught, and it can be taught at the school level with measurable results. The interventions that work share a common structure. They require students to make their thinking visible, and to articulate their plan before starting, to monitor their comprehension while working, and to evaluate their outcome against their original intention. This is not a natural way for most students to engage with academic work; it has to be explicitly taught and consistently reinforced.

Research on structured e-worksheets designed around problem-based learning frameworks found significant improvements in secondary students' metacognitive abilities across all three phases — planning, monitoring, and evaluating — after structured intervention, where students practiced articulating their reasoning and checking their understanding at regular intervals rather than working straight through to an answer (Mitasari and Hidayah, 2022). The percentage of students who reported being able to train metacognitive abilities through this kind of tool was high, but the critical mechanism was not the digital format; it was the requirement to stop and reflect before receiving feedback (Suarningtyas, 2022).

Goal-setting instruction is a particularly well-evidenced component of metacognitive training. Students who are explicitly taught to set specific, measurable learning goals — and who practice comparing their current understanding against those goals — develop better self-regulation over time (Faujiah et al., 2023). The research suggests that goal-setting works not just because goals provide direction, but because they create the comparison standard that metacognitive monitoring needs: *a student cannot notice that they are behind without a clear picture of where they intended to be*. This has a direct implication for how AI is used in study contexts. Frameworks that require students to articulate their own goal for an interaction before they prompt the AI (e.g. “I am trying to understand why the digestive system produces enzymes. I’m not sure what is an enzyme”) preserve the metacognitive comparison that passive consumption eliminates (Kong et al., 2024).

AI-augmented teaching, when designed carefully, can support metacognitive development rather than bypassing it. Studies on collaborative teaching models that combine instructor guidance with AI monitoring systems find that the students who benefit most are those whose AI interactions are structured around transparency — where the AI provides explicit feedback on the student's reasoning process (Yang and Xia, 2023). Deep Reinforcement Learning approaches to metacognitive tutoring offer an interesting glimpse at what this could look like at scale: systems that adapt their level of intervention to the specific type of metacognitive gap a student is showing, whether it is declarative (knowing about), procedural (knowing how), or conditional (knowing when and why) knowledge deficits, can help students with low incoming metacognitive knowledge outperform peers who had higher metacognitive knowledge but no structured intervention (Abdelshiheed et al., 2023). The pattern suggests that the relevant design question is not “how much AI support?” but “what kind of support, calibrated to which specific gap?”

What teachers do in this picture is not incidental. Self-regulation research on

teacher behavior finds that autonomy-supportive teaching is a significant predictor of students' ability to resist the kind of irrational avoidance that produces procrastination and disengagement in secondary school (Valenzuela et al., 2020). In this type of teaching strategy, the students are provided with reasons for tasks, acknowledging their difficulties, and supporting their goal-setting rather than simply directing it. The teacher's role in an AI-integrated classroom is not primarily to supervise AI use; it is to model and support the metacognitive habits that make AI use productive rather than substitutive. The research on teacher co-design of AI curricula is consistent with this: teachers who understand what metacognition is, and why it matters, are the ones better capable to design AI-integrated activities that require it (Van Brummelen and Lin, 2020).

The synthesis that emerges from the research on metacognition and self-regulated learning in young learners is the same synthesis that emerged from the age-specific risk analysis earlier in this chapter: the problem is not AI itself, but the conditions under which it is used. Used in ways that preserve the comparison signals, the productive struggle, and the explicit metacognitive reflection that accurate self-assessment requires, AI can genuinely support the development of self-regulated learners. Used in ways that eliminate all of that in the service of efficiency, it undermines the very capacities it is most often said to enhance.

Critical Thinking Dispositions Across Age Bands

The chapter so far has described the risks of AI to developing cognitive capacities, the mechanisms by which metacognition and self-regulation are undermined, and the evidence on what effective intervention looks like. There is one more dimension of the story that the preceding sections have not fully addressed: the difference between having the cognitive ability to think critically and being disposed to do it. This distinction turns out to matter enormously for how we understand what goes wrong when young people interact uncritically with AI outputs, and what it would actually take to fix it.

Disposition and Ability: Why the Distinction Matters

The dominant discourse around critical thinking in schools frames it primarily as a cognitive skill — something that can be assessed, trained, and measured. A student either can or cannot identify a logical fallacy, evaluate the credibility of a source, or construct a valid argument. This framing captures something real, but it misses an equally important dimension: the disposition to engage in these processes in the first place. We need to introduce two new concepts from the field of Cognitive Science and Educational Psychology: **epistemic curiosity** refers to the drive to seek new information and resolve uncertainty, whereas **intellectual humility** refers to the willingness to acknowledge the limits of what you know. These are not cognitive abilities, but rather motivational and affective orientations that determine whether a person reaches for their critical thinking tools at all (Usanova et al., 2024).

In early childhood, epistemic curiosity does not look like philosophical inquiry. It looks like the relentless “why?” of a five-year-old, or the way young children in structured STEM settings will return to a problem they could not solve, not because a teacher asked them to, but because the unsolved problem bothers them (Intisari et al., 2024). This natural drive is the engine of learning, and it operates before any formal instruction has occurred. **The educator’s task in early childhood is not to install curiosity**, but to avoid extinguishing it by replacing open questions with closed answers too soon.

Intellectual humility, our second complementary disposition, is more fragile and more dependent on the environment. Admitting that you do not know something requires a level of security that young children are still developing and that greatly depends upon the stability of a family structure that validates vulnerability. Research on the role of self-efficacy in learning motivation finds that students who believe in their capacity to learn are meaningfully more willing to engage in the vulnerable act of admitting ignorance, which is the prerequisite for actually filling the gap (Gania et al., 2023). When a student’s primary strategy for handling uncertainty is to pretend understanding, the metacognitive process stalls: there is no gap to fill if the student will not acknowledge the gap exists.

The development of intellectual humility is also tied to emotion regulation in ways that are not always recognized. The capacity for cognitive reappraisal — the ability to reframe a frustrating or threatening situation so that it becomes manageable — begins to emerge between ages three and five, and continues to develop through adolescence with increasing independence from adult scaffolding and supervision (Willner

et al., 2022). A child who cannot manage the frustration of not knowing something cannot sustain the intellectual humility that resolves the not-knowing; the emotional experience of confusion or failure overwhelms the epistemic process. Research on the co-development of behavioral symptoms and cognitive ability confirms this: children on an increasing trajectory of *externalizing symptoms* — expressing behaviors directed outward toward the environment like aggression, impulsivity or defiance — tend to show declining cognitive performance, not because their raw cognitive capacity has changed, but because emotional dysregulation has compromised their ability to deploy it (Papachristou et al., 2019). Critical thinking dispositions and emotional regulation are not parallel tracks; they run on the same rail.

This has a direct implication for AI use in educational settings. An AI tool that resolves uncertainty instantly, that answers the question before the child has sat with not knowing, is not just bypassing a cognitive process. It is bypassing the emotional and dispositional development that the experience of not knowing is supposed to catalyze. The curiosity never fires, and the intellectual humility never finds an opening to develop. Research on early reading habits finds that children who develop sustained engagement with texts from an early age show better cognitive outcomes and mental well-being in adolescence, and the proposed mechanism is similar: voluntary exposure to difficulty, ambiguity, and the gradual resolution of both (Sun et al., 2023). The capacity that matters in adolescence is built by the experiences that were not resolved for the child in childhood.

When Reasoning Skills Transfer and When They Do Not

The second insight this chapter’s manuscript adds concerns reasoning skill transfer — the question of whether critical thinking learned in one domain carries over to another. The intuitive assumption is that it does: a student who learns to reason carefully about evidence in science class should be better at evaluating evidence in history class. The research picture is more complicated.

The foundational components of logical reasoning — decomposition, pattern recognition, abstraction, and sequential planning — do develop through early computational thinking instruction in ways that appear to generalize (Ihsan et al., 2022). Students who have been explicitly taught to break problems into parts and look for patterns across instances show improved reasoning in contexts that share the same logical structure, even when the surface content is different (Pratama et al., 2024). This transfer is real, but it is not automatic. It requires the student to recognize that the structural similarity exists, which is itself a metacognitive operation: you have to step back from the specific problem and notice that it resembles something you have worked on before.

A persistent bottleneck in this transfer process is working memory capacity. When a problem becomes sufficiently complex, the cognitive load of managing its components can exceed the available capacity, and reasoning degrades not because the student lacks the skill but because the cognitive architecture is overloaded. Research on structured reasoning frameworks in demanding domains shows that scaffolds which reduce the organizational burden — such as a grid that separates factual analysis from inference from conclusion — can allow students to apply logical skills that would otherwise be swamped by the management of complexity (Burton, 2017). The scaffold does not bypass the reasoning; it reserves cognitive capacity for the reasoning by absorbing the

organizational work.

AI tools can, in principle, play the same role. Used well, they can hold part of the structure of a problem so that the student can focus on the reasoning. Used badly, they do the reasoning too, and the transfer that was supposed to happen does not — because what transfers across domains is not the product of reasoning but the practiced process of reasoning. When AI provides answers before a student has attempted to apply a familiar logical structure to a new situation, the moment of transfer, which requires the student to notice the structural similarity, is skipped.

Digital Epistemology: What Evaluating AI Outputs Actually Requires

The concept of digital epistemology refers to the skills involved in evaluating the credibility, accuracy, and underlying assumptions of digitally mediated information. For most of the history of digital literacy education, this meant evaluating websites: checking the author, the date, the domain, the citations. AI-generated content requires a genuinely different evaluative framework, and the differences are large enough that the old framework misleads more than it helps.

The most important shift is the one from evaluating *who* said something to evaluating *how* it was generated. A student assessing a Wikipedia article asks (or should ask): who wrote this, is that person reliable, when was it last updated? A student assessing an AI-generated response needs to ask instead: where does the training data that shaped this output come from? how would I know if it were confidently wrong? Am I being lied to or manipulated right now? These are harder questions, they are less intuitive, and they require a form of epistemic vigilance that does not develop automatically from general intelligence or from simple familiarity with AI tools.

The research on how students actually develop AI literacy documents a significant gap between usage competence and critical competence (Singh et al., 2026). Students who have been using AI tools regularly for academic work are often highly fluent users in the sense of knowing how to prompt effectively and how to work with AI outputs. They are often not reliably better at detecting when those outputs are wrong, biased, or fabricated. This confidence-competence paradox is particularly pronounced for AI because the outputs are designed to be fluent and authoritative: a hallucination in a large language model looks exactly like a true fact, formatted the same way and expressed with the same confident tone. There are no typographical cues or obvious uncertainty signals to trigger the student's evaluative response.

This effect plays out differently across age bands. Young children in early and middle childhood lack the metacognitive distance to recognize that a fluent, friendly response might be wrong; their developing *theory of mind* — the capacity to understand that other people have thoughts, beliefs, knowledge, and emotions that are different from your own — leads them to extend social trust to AI agents in a way that hurts critical evaluation before it begins (Intisari et al., 2024). Early adolescents face the additional complication of peer conformity: if an AI output aligns with what a peer group believes, the social pressure to accept it can override whatever evaluative disposition the student has developed (Papachristou et al., 2019). Late adolescents have greater capacity for the abstract skepticism that AI literacy requires, but they also have higher usage familiarity, which can produce the counterintuitive result that more experience with AI correlates with less critical engagement rather than more,

because confidence grows faster than competence.

The research on naturalistic AI literacy development suggests that what distinguishes genuinely critical AI users is not skepticism as a default attitude but something more specific: a set of “repair practices” that kick in when an AI output does not fit their prior knowledge or judgment (cha, 2026). Critical users notice breakdowns — moments when the AI response contradicts something they believe to be true, or produces a claim they cannot verify — and respond by re-prompting, cross-referencing, or explicitly asking the AI to justify its reasoning. Students who treat AI outputs as material to work with and evaluate, rather than conclusions to accept, develop these repair practices through use.

Instruction that builds digital epistemology therefore needs to do two things that standard AI literacy curricula typically skip. It needs to create situations where breakdowns occur — where the AI is demonstrably wrong about something the student can verify — so that the repair practice has occasion to develop. And it needs to address AI-specific forms of bias, particularly the way that large language models can systematically underrepresent certain perspectives or reinforce existing inequalities in ways that are invisible in the fluency of the output (Peer Mohamed, 2024). Students who have been taught to ask “whose perspective is missing from this response?” or “what assumptions does this output make about who the reader is?” are doing something that no checklist of AI literacy skills captures adequately. They are exercising the willingness to remain unsatisfied with a fluent answer until its epistemic status (i.e. its level of personal understanding) has been examined.

Assessment design is the final lever. When assessment tasks require only the kind of summary, report, or structured argument that AI can generate well, the evaluative skills described above have no practical context in which to develop. When assessment tasks embed the kind of contextual specificity, personal reflection, and disciplinary nuance that AI cannot replicate — that require the student’s own judgment about what is relevant and what is not — the student must engage their critical disposition to produce a satisfactory response (Gonsalves, 2025). The assessment design is not just measuring critical thinking; it is creating the conditions under which critical thinking has to be exercised. That is one of the oldest principles in education, and it applies to AI exactly as it applies to any other tool.

CHAPTER 4

The Home as a Learning Environment

The preceding chapter examined cognitive development and AI risk from the inside — looking at what happens to a child’s developing metacognition, self-regulation, and critical disposition when AI tools are used without deliberate design. This chapter moves outward, to the environment in which that development happens. For most children, the primary learning environment is not the classroom but the home. It is where the first questions get asked and where the relationship with technology first establish. Understanding what parents can actually do to shape this environment is the practical core of this chapter, and really, a central argument of the book’s argument.

The Home as a Learning Environment in the AI Era

For most of the twentieth century the home learning environment was defined by what it contained: the presence or absence of books, whether parents read to their children and dinner table conversations that involved the discussion of ideas. The research on these factors is extensive and consistent. What happens at the home in the early years predicts a remarkable share of a child’s later cognitive and academic trajectory, through pathways that run through language exposure, attachment security, and the modeling of intellectual habits. **A home that treated questions as worth pursuing tended to produce children who pursued questions.**

Generative AI has not changed the underlying developmental logic, but it has significantly complicated the landscape. The tools now available to a child sitting at a kitchen table are qualitatively different from anything that existed a decade ago (or even 5 years ago). A question that would previously have gone unanswered until the next school day, or prompted a trip to a reference book, or led a curious child down a productive path of partial knowledge and further wondering, now may receive an instant, fluent, confident, multi-paragraph response. Research on students’ relationship with AI tools finds that they increasingly treat generative AI not as an external aid but as essential infrastructure, comparable in their minds to internet access itself. This is, something the learning environment simply provides (Singh et al., 2026). The implication is significant: children are not choosing to use AI; they are navigating an environment in which AI is the default.

The opportunity the AI era creates is real but conditional. Used well, AI in the home can provide explanations in multiple formats, allow children to pursue their

curiosity at their own pace, and offer the kind of patient, non-judgmental interaction that is sometimes hard to sustain in a family under time pressure. What makes the difference is not which AI tool a family uses. It is whether the interaction preserves the child's role as the one who is thinking. AI used as a collaborator whose outputs require evaluation and editing is developmentally different from AI used as an oracle whose outputs are accepted and copied (Abdelghani et al., 2023). The same tool, the same question, the same response, but polar opposite developmental consequences, depending on what the child is asked to do next.

The Ecology of Reasoning: Space, Routine, and Climate

The home is not just a place where learning happens. It is a system of environments, rhythms, and relationships that either make sustained intellectual engagement more likely or less likely. Educators and developmental psychologists sometimes call this the *ecology of learning*: the idea that cognitive development is not just a property of the individual child but of the child in a context. For parents thinking about AI and reasoning habits, this ecological perspective is useful because it reframes the question. Instead of “how do I manage my child’s AI use?” the more productive question is **“what kind of environment am I building, and does it support the kind of thinking I want my child to develop?”**

Physical space matters more than is commonly appreciated. Homes that have areas dedicated to making and exploring — a table where things get spread out and worked on, a microscope or telescope, a bookshelf within reach, materials for building or drawing or experimenting — signal that intellectual activity is expected and valued. This is not about expense; it is about the symbolic architecture of the environment (a poster with a microscope can have a similar effect). A child whose default space is a bedroom with a large screen and no desk is in a different cognitive state. Research on home learning environments finds that children who grow up in settings where the physical environment supports interaction, play, and exploration show stronger self-efficacy and are more willing to engage with unfamiliar people and problems later (Li et al., 2023). The environment itself is doing pedagogical work.

Daily routine is the temporal version of this. Children who have predictable rhythms (regular mealtimes, a consistent homework time, bedtime that is actually bedtime) allocate cognitive resources differently than children in less structured environments, because they are not spending executive function on managing uncertainty about what comes next. The specific content of family routines matters less than their existence and their quality. Research on intensive parenting frameworks finds that the moments families most value are not formal “educational” interactions but the togetherness embedded in ordinary shared activities (a meal, car ride, or a walk) when the child’s mindset is in a passive state relative to learning. During this time conversations happen naturally (Kutrovátz, 2022a). A parent who uses the drive to school to ask what the child is working on, or to share something they found puzzling, is doing more for the child’s critical thinking development than a parent who schedules a separate “learning time” but otherwise defaults to screens. We need to stress this enough: **the change begins with the parent, not the child.**

The socioemotional climate is the most powerful variable of the three, and the hardest to change. Children develop the willingness to say “I don’t know,” to try something and fail or revise a view that turned out to be wrong (i.e. intellectual courage) in environments where those things are safe. When the pressure is on correct answers, on performance, on not wasting time, children learn to perform understanding rather than actually seek it. They find shortcuts. AI is an extraordinarily efficient shortcut, and a child who has learned that appearing to know is more rewarded than actually figuring things out will use it without hesitation, unless critical thinking is a family value. What’s more, the temptation will remain as a potential push back throughout the child’s development and adulthood, as this pattern of “product at the expense of understanding” is a short term reward in traditional education (scores), social media (likes) and businesses (money), but as we mentioned in previous chapters, there is no such thing as free lunch, and often the lack of “vision” comes from the fallacy of mistaking the metric for the target, and revealing a lack of critical thinking along the way.

Research on parental educational beliefs and home learning environments finds that children whose parents emphasize understanding and process over performance are more likely to develop the self-regulatory capacities that underpin deep learning (Li et al., 2023). Supporting a child’s sense of autonomy — allowing them to approach problems in their own way, tolerating inefficiency, not immediately correcting — predicts better self-regulation outcomes than directive guidance, even when the directive guidance is well-intentioned (Valenzuela et al., 2020). The socioemotional climate that supports reasoning is one where thinking out loud is normal, where not knowing something is the beginning of a conversation rather than an embarrassment, and where the adults in the household are visibly engaged in their own intellectual lives.

The emotional dimension of AI interaction specifically deserves mention here. Children who use conversational AI tools often develop something resembling a relationship with them: they express gratitude when the AI helps, frustration when it fails, and sometimes genuine anxiety about whether they are using it “correctly” (Cha, 2026). This is not a trivial phenomenon. It represents the social cognition that children naturally apply to any responsive agent being activated by a system that is designed to be helpful and friendly. A home climate that normalizes critical engagement with AI where it is entirely acceptable to say “the AI got that wrong” or “I don’t trust that answer, let’s check”, is doing something genuinely significant. It is decoupling the child’s social dynamics from the interaction before they get normalized. This is also a matter of safety, as some models have famously misbehaved and even encourage self-harm, resulting in tragedy.

Parental Mediation: From Restriction to Active Engagement

The research on how parents actually mediate their children’s technology use has been building for decades, first in the context of television, then video games, then social media. The consistent finding is that the most effective form of parental involvement is neither restriction nor permissiveness: it is active mediation — co-engaging with

the child around the content, discussing what is being seen or read, asking questions that prompt evaluation (Zaman et al., 2016). Restriction alone tends to produce children who find workarounds; unmediated access tends to produce children who develop uncritical habits. The middle path, which requires the most parental effort, produces the best outcomes for critical thinking development.

The application to AI is direct, and it maps cleanly onto the developmental concerns the previous chapter described. A parent who simply monitors screen time, or who sets content filters, or who bans AI use for homework, is practicing restrictive mediation. These strategies address symptoms rather than developing capacity. A parent who sits with the child and asks “what did the AI say about that?” and “does that match what you already knew?” and “how would you find out if it’s wrong?” is doing something categorically different. They are keeping the evaluative work with the child while using the AI interaction as a resource.

Research on the factors that shape parental mediation quality shows a significant gradient by parental education and digital confidence (Alkan and Mertkan, 2021). Parents with higher levels of education and greater comfort with technology are more likely to engage in active mediation, guiding their children toward critical evaluation. Parents with less confidence tend toward restriction or passive acceptance, either blocking AI use or allowing it without engagement. The consequences compound: a child who receives active mediation develops evaluative habits that generalize; a child who receives only restriction develops neither the habit nor, once the restriction lifts, the tools to manage without it. This gradient is arguably a more fundamental source of educational inequality in the AI era than differential access to premium AI tools, because mediation quality shapes capacity, whereas tool access shapes only opportunity (Kutrovátz, 2022b).

The most direct form of active mediation is guided questioning. The principle is old — Socratic dialogue, philosophical inquiry in schools — but its application at home is straightforward. A parent does not need expertise in AI to ask the right questions. “How do you know that?” “What would change your mind about that?” “Who might disagree with that, and why?” “What is the AI leaving out?” These questions push the metacognitive work back to the child. They require the child to treat the AI’s output as material to evaluate rather than a conclusion to accept. Research on philosophy education programs that train children in exactly this kind of dialogic evaluation finds that students who engage in structured inquiry — articulating reasons, responding to challenges, reconsidering their positions — develop significantly (in the statistical sense) stronger critical thinking skills than those who do not, across age groups and disciplines (Faraji et al., 2018).

Parental modeling is the quieter mechanism, but it may be more powerful than explicit instruction. Children watch how adults navigate uncertainty. A parent who, when faced with a claim they are unsure about, says “I don’t know; let me think about that” or “that doesn’t seem right to me; let me look for another source” is demonstrating in real time what critical engagement looks like. A parent who asks the AI for the answer and accepts whatever comes back is modeling the opposite. The research on scientific thinking consistently emphasizes that the disposition toward skepticism and verification is caught as much as it is taught. This means that it develops through exposure to adults who practice it visibly (Schmaltz et al., 2017).

One of the most effective things a parent can do for a child’s AI literacy may simply be to practice their own.

Finally, mediation needs to be developmentally responsive. What active mediation looks like for a seven-year-old is different from what it looks like for a fifteen-year-old. For younger children, the focus is on the most basic epistemic question: is this true? Does this match what we know? For older children, the questions can expand to include: how was this generated? What training data shaped this? Whose perspective is absent? What would this look like if it were wrong? Research on adolescent cognitive control development shows that the capacity for this kind of abstract metacognitive questioning — thinking about the conditions under which a source might be unreliable — matures significantly in the teenage years as prefrontal functions consolidate (Crone and Steinbeis, 2017). The parental mediation that serves a teenager well builds on the earlier, more concrete forms rather than replacing them: by adolescence, the goal is not to model skepticism but to support the teenager’s development of their own.

Parental Mediation of Digital Technology

The previous section established that active parental engagement through guided questioning, modeling of skepticism and age-appropriate supporting tools is more effective than restriction alone. But framing mediation as simply “active versus restrictive” misses the genuine complexity of what parents may encounter, and why. Research on parental mediation has built substantial progress over several decades, initially in the context of television and social media. We can use some of those strategies as a starting point to address the mediated use of AI tools. Understanding these distinctions matters because different strategies produce different outcomes depending on the child’s age, the type of technology, and the family’s circumstances. This section goes deeper into the mechanics of such classification, examines what the evidence actually shows about screen time, and addresses the specific ways that algorithmic and generative AI environments break the assumptions on which traditional mediation was built.

The Classification of Strategies and When Each Approach Works

Contemporary mediation research recognizes four distinct strategies that parents deploy, often in combination: active mediation, restrictive mediation, monitoring, and deference (Nagy, 2022). These are not a hierarchy; none is universally superior, and the effectiveness of each depends significantly on the developmental stage of the child and the nature of the technology being mediated.

Active mediation involves engaging with children’s media use through discussion, co-viewing, and guided questioning. This is the approach most associated with positive cognitive and social outcomes. When parents interpret and question content alongside their children, they provide an external critical framework that the child gradually internalizes (Zaman et al., 2016). Crucially, the distinction between passive co-use (being in the room while the child uses a device) and active co-use (engaged discussion about what is happening on the screen) matters enormously: the former has negligible or even negative effects, while the latter transforms media consumption into a dialogic, literacy-building activity (Seon Pek, 2025). A parent who sits next to a child watching a YouTube video and says nothing is not mediating; a parent who asks “why do you think the video showed it that way?” or “does that seem accurate to you?” is. Find common areas of interest by exploring the internet or asking questions together, and be comfortable acknowledging your learning as an adult in the room.

Restrictive mediation is represented by time limits, content filters, access rules, etc. This type of strategy is consistently effective during the preschool years and becomes progressively less effective, and occasionally counterproductive, as children age (Fitzpatrick et al., 2023). For very young children, externally imposed limits on screen exposure help regulate behaviour before self-regulation capacity has developed; pre-established routines work better than reactive restrictions. For adolescents, the picture is nearly reversed. Research on teenagers’ responses to parental technology governance finds that adolescents frequently experience restrictive mediation as a signal of distrust, and that this experience has more influence on their willingness to discuss digital experiences with parents than any specific rule does (Adorjan and Ricciardelli, 2022). Teenagers whose parents rely primarily on restriction are not necessarily using technology less; they are using it more privately and discussing it less. The rule

produces concealment rather than compliance.

Monitoring means checking browsing histories, reading messages, tracking device use — sits in an uncomfortable middle ground. It offers parents visibility into their children's digital lives, but many parents themselves report discomfort with it because it conflicts with their values around trust and autonomy, particularly for older children (Nagy, 2022). Where monitoring is experienced by the child as surveillance rather than oversight, it tends to degrade the quality of the relationship that active mediation depends on.

Deference means granting children substantial autonomy over their own digital use, intervening minimally. This is not mediation so much as the absence of it, and in early childhood it tends to produce the outcomes one would expect from unsupervised exposure. For older adolescents with well-developed self-regulatory capacities, some degree of deference is appropriate and even necessary: the goal of mediation is ultimately to produce a young person who does not need it.

The socioeconomic gradient in which strategy parents actually use is one of the more remarkable findings in this literature. Parents with higher levels of education and greater confidence with technology are more likely to implement active mediation consistently; parents with less educational background or lower digital self-efficacy tend toward restrictive strategies, often because active mediation requires a level of comfort with the technology that they have not developed (Alkan and Mertkan, 2021). The consequence is that the children who would benefit most from active mediation — those in lower-resourced households, where other cognitive enrichment inputs may also be scarcer — are the least likely to receive it, not because their parents care less, but because active mediation requires knowledge and confidence that are themselves unequally distributed (Kutrovátz, 2022b). This is not a problem that content filters or parental control software solve. It seems evident that open-source and free resources for adults willing to make such investment should be a priority in the development of education software.

What the Screen Time Evidence Actually Shows

The popular discourse around screen time tends toward a simple narrative: less is better, more is harmful. The research is considerably more complicated. What matters is not primarily the quantity of screen time but its quality, context, and the degree to which it displaces or complements other developmental activities.

For very young children, the concerns are real but specific. Passive media consumption during early childhood — particularly when it displaces face-to-face interaction with caregivers — is associated with reduced opportunities for the contingent, back-and-forth exchanges through which vocabulary and social understanding develop (Palavan Özcan, 2025). This is not because screens are inherently harmful; it is because the things children are not doing while they watch passively — being talked to, asked questions, prompted to respond — are genuinely important for language acquisition and social-emotional development. Research on screen time and language development emphasizes that socioeconomic context matters significantly: the effects of screen exposure on language outcomes are not uniform across social groups, and guidance that ignores this heterogeneity is unlikely to serve families well (Seon Pek, 2025).

For school-age children, device placement within the home is a more consequential variable than most parents recognise. Research on behavioral correlates of screen time consistently finds that bedroom use is associated with more problematic patterns — reduced sleep, decreased attention, greater likelihood of exposure to content without parental awareness — compared to use in shared family spaces (Tarkhoun et al., 2026). The living room creates a degree of ambient social oversight without requiring active surveillance; the bedroom removes it. A policy of keeping devices out of bedrooms is one of the few straightforward structural interventions that research supports with reasonable consistency.

The more difficult finding is that the content and interactivity of screen use matter more than the clock. A child who spends two hours doing creative work with digital tools, or engaged in active collaborative play via a game with social elements, is in a different developmental situation from one who spends two hours passively watching auto-playing videos. The problem with the quantity focus in public guidance is that it flattens this distinction, making parents anxious about a number rather than attentive to what the number is made of. Parental perceptions of digital media involve genuine ambivalence: most parents recognise that digital tools offer real developmental opportunities even as they worry about the risks (Palavan Özcan, 2025). The most useful reframe is from screen time management to interaction quality management — asking not “how long?” but “what is the child doing, with whom, and with what adult involvement?”

When Algorithms Change the Mediation Problem

Traditional parental mediation was designed for a media environment with relatively stable and identifiable content consisting of television programs that had titles, runtime, and ratings. Parents could watch what their children watched, assess it, discuss it. Algorithmic recommendation systems have broken each of those assumptions.

Modern video and social media platforms serve content generated in real time by systems that observe granular behavioural signals — watch duration, scroll speed, pause patterns, revisits — and optimise for continued engagement. The content a child encounters is not selected from a fixed catalogue by the child; it is dynamically generated by a system that learns to show whatever keeps the child watching. Parents who practise traditional active mediation (i.e. watching alongside the child and discussing what appears) are co-watching something that neither parent nor child chose and that will be different the next time the child uses the platform. This is a categorically different mediation problem from discussing a television show.

For generative AI tools specifically, the challenge is different again. Research exploring parent and child perceptions of safety in generative AI environments finds that parents’ primary concerns cluster around two issues: the accuracy of AI-generated content and the opacity of the system’s reasoning (Yaman Yu, 2024). Children’s concerns, elicited separately in the same research, are more relational — they worry about the AI providing content that is hurtful or socially inappropriate, and they want to know when they can trust what the AI says. Both groups agree that they want transparency about what the AI is doing and why. Neither group currently receives it reliably.

What this research suggests is that effective parental mediation of AI tools requires parents to understand something that even technically sophisticated adults often do

not: that generative AI systems produce outputs that are calibrated to be plausible and fluent, not necessarily true, and that the system's confidence in its output is not a reliable signal of its accuracy (Singh et al., 2026). Children, whose developing source-evaluation skills lead them to use fluency and confidence as proxies for reliability, are particularly vulnerable to this mismatch. A parent who understands this can make it explicit: "the AI gave a very confident-sounding answer, but that's not the same as it being right. How would we check?" A parent who does not understand it will not know to ask. When Google launched its AI-generated search summaries in May 2024, users experimented with unusual queries. There are some famous cases out there. When asked how many rocks a person should eat, the AI confidently generated a response claiming that according to UC Berkeley geologists, eating pebbles or gravel was an excellent source of vitamins and minerals. In other instance, the reliance on species identification of poisonous mushrooms labelled as "eatable" by generative AI led to several hospitalizations and even deaths.

The privacy dimension requires specific attention. AI tools used by children in educational and recreational contexts collect data about the child's questions, interests, confusions, and knowledge gaps. The appropriate framework for handling this data is meaningfully different from the framework for adult users, and most current generative AI products were not designed with child data protection as a primary constraint (pri, 2026). Parents cannot currently inspect or control what AI tools remember across sessions, what is logged, or how interaction data might be used. This is a structural gap that requires both better product design and better regulatory frameworks; in the interim, parents are well-advised to treat AI tools as unaudited data collectors and to prefer products that make explicit commitments about session isolation and data minimisation for younger users.

The shift this implies for parental mediation is significant but navigable. The parent's role is not to become a technical expert in machine learning (although it's never late to get into it!). The goal is to maintain an active, critical presence in their child's AI interactions, not by monitoring every prompt, but by ensuring that AI use is not fully privatised, by modelling the habit of questioning AI outputs, and by treating "how do we know this is right?" as a routine question rather than a sceptical exception. The children who develop genuine AI literacy, as opposed to mere AI fluency, are the ones for whom that question is a reflex rather than an interruption.

Homework, AI Assistance, and Parental Oversight

The sections of this chapter so far have described the home learning environment in broad terms: what the physical space and routine afford, how parental mediation operates, what the screen time evidence actually shows, and how algorithmic systems complicate traditional oversight. This section narrows to a specific and practically urgent question: when a child sits down to do homework with a generative AI tool available, what is actually happening, and what can a parent usefully do about it?

The Tutoring–Ghostwriting Spectrum

One of the more intellectually honest things the research on AI in homework has established is that the line between legitimate AI assistance and academic outsourcing is genuinely not easy to draw, and that students know this. Research on student perceptions of AI misconduct finds that many students do not experience themselves as cheating when they use AI for homework; they experience themselves as using available tools, much as they might use a calculator or a spell-checker (Lund et al., 2024). The genuinely difficult question is why using a grammar checker is acceptable, using a dictionary is acceptable, using a search engine is acceptable, but using a generative AI to draft a paragraph is not. The answer to that question has to go deeper than “because the policy says so” if it is going to mean anything to a child in practice (Evangelista et al., 2025).

Unsurprisingly, the most useful framework here is not binary (AI is permitted or forbidden) but a continuum organized by who is doing the intellectual work. At one end, a student who has written a draft, read it critically, identified where an argument is weak, and used AI to suggest better phrasing for a specific sentence is in a fundamentally different cognitive relationship with the task than a student who has typed the assignment prompt into a chatbot, read the output, and submitted it with light editing.

Research on student AI use in writing-intensive tasks identifies the cognitive signal that marks the difference: the presence or absence of an authorial stance. Students who use AI strategically maintain a clear evaluative relationship with the output — they know what they think, they use the AI to express or refine it, and they can defend their argument without the AI present. Students in the passive-reliance mode gradually lose this relationship; their work begins to exhibit what researchers describe as depersonalisation, characterized by a reduction in first-person reasoning markers and an increase in hedged, non-committal language that was not there before (Zafar, 2026). In a more professional setup Heavy AI-mediated writing is identifiable not by its quality, which tends to be high, but by its absence of a particular intellectual presence — the sense of a mind working out what it thinks.

Academic integrity frameworks that try to address this at the institutional level increasingly recognize that detection-based approaches (AI detectors, plagiarism scanners) are structurally inadequate. No detection system can reliably distinguish good AI-assisted writing from good human writing, and students quickly learn this (Pokotylo, 2025). The more productive shift, at both the institutional and the home level, is toward process-based evidence: portfolios, annotated drafts, reflective commentaries that document how a student arrived at a conclusion, and oral explanations of the work

(Dau et al., 2026). A student who can explain why they made the argument they made, what they considered and rejected, and what they are still uncertain about is demonstrating intellectual ownership regardless of what tools they used to produce the final text.

Who Becomes Dependent and Why

The research on AI dependency in student populations has moved beyond the question of whether dependency occurs to the more useful question of what predicts it. A latent class analysis of student AI use found distinct profiles: a group of strategic users who engaged AI as a supplementary resource while maintaining control of the core reasoning, and a group characterized by substitutive reliance, where AI was used as a primary generator of academic content, with students functioning more as editors than as authors (Fernando et al., 2025). The substitutive group showed the most significant declines in self-assessed critical thinking, research, and writing competence over time, even though the amount of time spent solving the problem was comparable.

The mechanism underlying this divergence appears to be prior domain knowledge. Research distinguishes two relationships a learner can have with an AI tool: the prosthetic relationship, in which the learner has enough foundational knowledge to evaluate, direct, and critically engage with AI outputs; and the crutch relationship, in which the learner lacks that foundation and therefore has no alternative towards AI outputs but to accept them (Khan, 2025). The implication for homework contexts is counterintuitive: the students most likely to benefit from AI assistance in the short term are precisely the students who need it least. Students with strong prior knowledge use AI to extend what they can already do; students with weak prior knowledge use it to avoid doing the thing that would develop the knowledge they lack (Amir Yunus, 2024). This creates a dynamic in which the performance gap between stronger and “weaker” students narrows in the short term, where weaker students produce better-looking work with AI help. In contrast, the competence gap between the two groups widens because the stronger students are developing through use and the weaker students are not. This is one of the more troubling structural features of AI in homework, and it is not addressed by policies that restrict AI equally across all students. What is more, failure from institutions insisting on retaining a pure metric system to evaluate their student’s learning can produce unskilled professionals.

Generative AI interacts with this dynamic through metacognitive shortcuts, which I’d argue is still a concern to skilled professionals as it may hide semantic issues related to the information domain of the problem they’re trying to solve (always remember to decompose the problem before prompting!). When a student can prompt an AI to produce a plausible answer, they bypass the signal that motivates learning (i.e. the epistemic curiosity we discussed in Chapter 3). Research on this pattern describes it as the elimination of curiosity-driven information search: rather than detecting a knowledge gap and pursuing it, the student asks the AI, receives a confident answer, and moves on without the gap ever becoming visible (Abdelghani et al., 2023). The downstream consequence is that the student’s model of their own understanding becomes progressively less accurate, affecting their intellectual humility. They believe that they know more than they do, because they have not experienced the situations that would reveal the discrepancy (Banerjee, 2026).

Practical Guidance Frameworks for Home Study

What can I do, as a parent and educator? This parental guidance question that follows from all of the above is how to ensure that the use is prosthetic rather than substitutive. This requires interventions that preserve the cognitive work the homework was designed to require.

The most important structural principle is that AI assistance should come after the student has attempted the work independently, not before. This is not a punitive rule; it is a developmental one. The attempt — even a failed, incomplete, or confused attempt — establishes the cognitive state that makes AI assistance useful. A student who has tried to write a paragraph about the causes of a historical event and found their argument trailing off has activated the question of what the argument should actually be. When they then engage with AI to refine or challenge their draft, they are in a position to evaluate the AI's suggestions against their own developing view. A student who goes to the AI first never develops that view at all (sma, 2026).

Parents can make this concrete by requiring a “pre-AI draft” before any AI interaction for writing tasks — not a finished piece, but whatever the student can produce on their own. For problem-solving tasks, this means attempting the problem and identifying specifically where the approach breaks down before consulting AI. This practice is not about creating extra work; it is about ensuring that the AI interaction is diagnostic rather than generative — that the student is asking “does the AI agree with my reasoning?” rather than “what is the reasoning?” (Dranias and Whitley, 2026).

When students do get stuck — genuinely stuck, not merely uncomfortable — the quality of the intervention matters as much as its timing. Research on tutoring systems consistently shows that providing direct answers produces worse learning outcomes than providing scaffolded guidance: explaining, reframing, asking a question that points toward the issue, or exemplifying without solving (Liu et al., 2024). Parents do not need to know the subject matter to apply this principle. “What have you tried so far?”, “What part of this do you not understand?”, and “What would you need to know to get unstuck?” are more useful than “let me see the question” followed by an explanation of the answer. AI tools, when used in a Socratic rather than oracular mode, can play the same role: asking the AI to explain why an approach failed, or to ask the student a question, rather than asking it to provide the solution (Lee et al., 2025).

Here, I'd like to provide a personal note. For older students, especially young adults, the Socratic method can sometimes feel patronizing and unrewarding, making active empathy essential to identify when the student is directly or indirectly communicating their need for a concrete answer from which the dialogue can move forward, at which point the educator should follow the student and adapt, especially if signs of frustration begin to show. I experienced this myself as a grad student when asking my supervisor a question, and expecting him to be explicit. My lack of experience communicating my own learning strategies and my supervisor's passive pedagogical voice made a terrible match that led me to months of feeling stuck and overwhelmed in academia. For a student who is trying to be acknowledged as an adult, this method can be detrimental if pushed too far.

Scaffolding, whether provided by a parent or an AI tool, should also be temporary. The goal is not permanent support but the gradual internalization of the capacity the

support was providing. This is what pedagogists call fading: as a student becomes more familiar with the cognitive demands of a task type the scaffold is progressively reduced until it is no longer needed. A child who needed a parent to prompt the pre-AI draft check every session eventually does it automatically; a child who is still receiving the same level of prompting a year later has become dependent on the prompt rather than on the habit (Burton, 2017). Parents should track whether their guidance is decreasing over time. If it is not, the intervention is maintaining performance rather than building capacity.

Strategic friction is the most active form of parental involvement in AI-assisted study. Rather than correcting mistakes or providing explanations, a parent introduces a deliberate challenge: a counterexample, a question the student did not consider, or an AI-generated response that contains a subtle error and asking the child to find it. This transforms the homework session from a production process into a diagnostic one. The student is not just completing the task; they are developing the evaluative instinct that will eventually allow them to do this themselves, unprompted, with every AI interaction they have (Dranias and Whitley, 2026).

Building the AI-Resilient Learner

The synthesis of everything in this chapter points toward a specific set of **habits** that distinguish learners who develop through AI use from those who are gradually replaced by it.

The first is *metacognitive vigilance*: the habit of noticing when understanding is genuine versus when it only feels genuine because a fluent answer is nearby. This is not a natural tendency. Fluency is persuasive, and even careful thinkers can mistake it for comprehension (an attribute cleverly exploited by oratory and politicians). What develops metacognitive vigilance is the repeated experience of checking: asking yourself whether you could explain this without the AI, whether you could apply it to a different problem, whether the argument would hold if the AI's specific examples were removed (Abdelghani et al., 2023). Parents can build this habit by making it a closing ritual for any AI-assisted homework session: “tell me the main idea in your own words, without looking.”

The second is *orchestration skill*: the ability to decompose a complex task, identify which components genuinely benefit from AI assistance and which require independent thought, and make deliberate decisions about when to delegate and when not to (Bhati, 2026). This is the disposition that distinguishes augmented learners from automated ones — not a general suspicion of AI, but a clear and developing sense of where their own thinking is valuable and where a tool can legitimately extend it.

The third is *process discipline*: maintaining a consistent structure for AI interaction that preserves the metacognitive sequence the task was designed to require. The 6-P pedagogical framework offers a practical model: plan the content independently, prompt the AI, preview and validate the output, produce the work, engage in reflection, and track progress over time (Kong et al., 2024). In a homework context, this can be adapted to age and subject, but the core structure — think first, then engage, then evaluate, then reflect — remains developmental regardless of the specific task.

None of these habits develops from instruction alone. They develop through prac-

tice in conditions where they are required — where the tools are available, the task is real, and an adult is present not to police or to help, but to ensure that the student remains the one doing the thinking. That is, in the end, the simplest description of what effective parental oversight of AI-assisted homework looks like: not the management of a tool, but the maintenance of a human being who is growing rather than being replaced.

Equity and Access in the Home Learning Environment

The old conversation about the digital divide had a certain clarity to it. A child either had a computer at home or did not. The school either had functioning computers in the library or it did not. These gaps were real and consequential, but they were at least legible—measurable in device counts and download speeds. Close the hardware gap. Connect the unconnected. The goals were hard to achieve and unevenly funded, but not hard to state.

That clarity is gone. As artificial intelligence tools become embedded in the daily work of studying, writing, researching, and preparing for exams, the nature of educational disadvantage has quietly shifted. We have moved from a question of access—*do you have the tools?*—to a question of quality and capability: *how powerful are your tools, and who decides which tier of help you can afford?* Researchers describe this transition as a move from the first-level digital divide, which was measured in hardware and connectivity, to what is now being called the *AI divide*, pointing at a disparity rooted not in whether a student can open an AI tool, but in how capable that tool is and whether the student has the skills to use it well (Singh et al., 2026).

This is not a subtle semantic shift. It changes which families are advantaged, which institutions need to respond, and what we even mean when we say a student has a fair chance.

The Entrance Fee

The use of AI tools at the level where they genuinely support complex academic work requires a reasonably capable device and stable connectivity. When these conditions are not met, the experience degrades in ways that compound over a homework session: slow response times, dropped connections, interfaces that load incompletely on older hardware, and the kind of persistent low-level friction that makes sustained, iterative work feel more effortful than it is worth (Chetry, 2024).

None of this means a student needs a high-end workstation to open an AI chatbot. Most of the tools students encounter are cloud-based and run on modest hardware. But modest is not nothing. The quality of the interaction scales with the quality of the device and connection in ways that matter even when they are not dramatically visible (Chetry, 2024). For families sharing a single device between multiple children, or managing older phones on limited data plans, this baseline is already a real constraint—and it operates entirely beneath the level at which school policy usually engages with equity. In the summer of 2026, at the time when this book was written, we are experiencing a short supply and a disproportionate increase in the pricing of essential components needed to build computers, caused by the high demand of resources for future data centers. This pattern promises to continue, or at least last, by the end of 2027, putting tremendous financial stress over families needing to buy digital systems.

The Freemium Problem

The central economic mechanism of the AI divide is the freemium model. Most leading AI providers offer a free tier and a paid tier. The free tier provides access to the tool, whereas the paid tier provides access to a meaningfully better version of it. Paid model

versions offer more sophisticated reasoning, a larger context window, and output that is less prone to confident errors(Singh et al., 2026).

Think of it like a gym where the basic membership gives you access to the floor and 5 pound dumbbells, but the equipment that actually builds strength is locked behind the premium plan. Everyone can technically say they use the gym but the experience and the results are not the same.

In an educational context, this distinction carries real weight. Premium AI tools are genuinely better at helping students work through complex problems, identify weaknesses in their arguments, and produce output calibrated to specific prompts. A student using a more capable AI assistant gets more reliable feedback, catches more errors in their reasoning, and receives responses that are actually responsive to the nuance of the question. A student on the free tier may get output that is fluent and authoritative-sounding but factually unreliable or analytically shallow—what researchers in this space have taken to calling *hallucinations*(Singh et al., 2026; Chetry, 2024).

When more capable reasoning is available only to students whose families can pay monthly subscription fees, what is being sold is not a convenience but a cognitive advantage—and one that compounds across a school year(Singh et al., 2026). Researchers have begun using the term *algorithmic paywall* to describe this condition: a regime in which the quality of a student’s intellectual assistance is determined less by what they know or what they are trying to learn than by their household’s relationship with a subscription billing cycle(Singh et al., 2026). The result is a potential bifurcation of the educational experience along economic lines that is invisible in any official record but visible in outcomes over time(Singh et al., 2026; Rodrigues, 2024).

Personally, I believe that open source AI models like DeepSeek and GLM 5.2 can help narrow these gaps, as these models offer much cheaper tiers and run much more efficiently in terms of tokens per request. The innovation behind these models relies on a technique called *distillation*, where the systems learn from more capable models. Said in other words, the notion of a premium-tier model leaves a gap in education (and a polarizing effect) that open source models can help narrow.

What Institutions Are Not Doing

One might expect schools and universities to step in here. If capable AI tools are becoming genuinely important to academic work, the argument for institutions to provide licensed access is structurally identical to the argument for library databases, writing center support, or course management software: these are educational resources, and access to them should not be contingent on ability to pay.

The evidence suggests that institutions have been slow to meet this expectation—and that, in the meantime, their primary response to AI has been restriction rather than provision. Students report a significant gap between the AI policies they encounter, which tend to be framed in disciplinary terms, and the practical, licensed support needed to engage with these tools effectively and fairly(Singh et al., 2026). Universities communicate rules about AI but have been considerably less effective at providing students with the resources to follow those rules from an equal starting position(Singh et al., 2026).

When an institution declines to provide licensed access to capable AI tools, it does

not neutralize the advantage held by students whose families can afford them independently. It simply shifts the cost of bridging that gap onto individual households(Chetry, 2024). For families with the means to subscribe, this is an inconvenience. For families without, it is a new and specific form of educational exclusion—one that operates invisibly, because it takes place outside the formal structures of tuition, curriculum, and grading. No committee voted to give wealthier students better AI tutors. It emerged from the structure of the market, and it will not be resolved without deliberate institutional action(Singh et al., 2026; Chadha, 2024).

How the Gaps Compound

These deficits in device quality and access to capable tools do not sit beside each other independently. They compound alongside the patterns outlined in our previous sections on how parental supervision is applied. A student managing an older shared device, working with a free-tier AI whose output is less reliable, and without a parent who can model how to interrogate that output critically, is not facing one disadvantage but three, and each amplifies the others(Singh et al., 2026; Zafar, 2026; Chetry, 2024).

Writers about this pattern have reached for the language of the *Matthew Effect*—the social dynamic, named for a line in the Gospel, in which those who already have advantages tend to accumulate more of them. In educational AI, the mechanism is precise: students from well-resourced backgrounds access premium tools, receive higher-quality output, have parents who can guide them toward strategic use, and develop stronger AI literacy as a result—which further increases the value they extract from the tools, which further widens the gap from peers who never had the same starting conditions(Singh et al., 2026; Chetry, 2024). The technology that was supposed to personalize learning and reach students that standardized approaches had always missed has become, in its current market configuration, a mechanism for reinforcing the hierarchies it was described as disrupting(Chadha, 2024; Rodrigues, 2024).

Long Shadows on the Labor Market

These gaps in the home learning environment do not dissolve at graduation. They extend forward into professional life in ways that are already becoming visible.

The labor market is shifting toward roles that reward the capacity to direct, evaluate, and work alongside AI systems. These are positions that require not just technical familiarity but the kind of higher-order critical judgment that only develops through genuine intellectual engagement(Ejjami, 2024). Students who spent their formative academic years using AI as a bypass rather than a scaffold are likely to enter this market poorly prepared for the roles that demand exactly the habits they practiced avoiding(Koch, 2026b; Ejjami, 2024).

The anxiety that students are already expressing about this is audible in how they discuss the future of their professions: whether degrees will retain their value, whether specific careers are “AI-proof,” whether the skills they are developing will remain in demand(Yüce et al., 2026). These concerns are not irrational. The students who will navigate this transition most effectively are those who developed genuine analytical capacities during their education—not because AI becomes irrelevant to their work, but because they will be equipped to direct it strategically rather than be displaced

by it(Ejjami, 2024; Koch, 2026b).

There is also a subtler layer of risk embedded in the tools themselves. AI systems used in language education, for instance, have been documented to encode assumptions about linguistic prestige that disadvantage students whose primary language or dialect does not map onto the dominant register the system was trained on(Peer Mohamed, 2024). When a student receives AI feedback that consistently marginalizes the way they naturally write or speak, and they lack both the critical literacy to identify this bias and the teacher support to contextualize it, they may internalize the AI's judgment as a personal deficit rather than a structural one(Peer Mohamed, 2024). The tool that was supposed to offer everyone the same quality of feedback ends up reproducing the same hierarchies that structured educational inequity before AI existed.

What Would Actually Help

Closing the AI divide requires engagement at all three levels simultaneously. Addressing one without the others produces at best partial progress and at worst a false reassurance that the problem is being managed when it is not.

On the material and financial level, institutions need to begin treating licensed access to capable AI tools as a question of educational infrastructure—not a luxury feature but a basic provision on the same footing as library holdings or writing support services(Singh et al., 2026; Chadha, 2024). The current default, in which this cost falls to individual families, is a policy choice with predictable distributional consequences. It will not resolve itself.

On the pedagogical level, AI literacy—including the metacognitive habits that distinguish strategic from passive use—needs to become a substantive and assessed part of the curriculum rather than an optional orientation session(Abdelghani et al., 2023). How to interrogate AI output. How to use AI to develop an argument rather than outsource one. How to recognize when a fluent-sounding response is also an unreliable one. These are teachable skills(Zafar, 2026; Chetry, 2024), and for the many students whose home environments cannot model them, the classroom may be the only reliable place to learn them.

The home learning environment sits at the intersection of all three layers of this divide. The advantages and disadvantages children carry into it do not arise from individual choices but from structural conditions—market structures, institutional defaults, and even geography that determines which parents have AI in their workflows and which do not. Changing educational outcomes on this front will require responses at the structural level, not just the personal one. That observation may feel discouraging to a parent reading this. But it points towards what individual families can experiment with at incremental, gradual changes. The gains are very much worth it.

CHAPTER 5

The Classroom in Transition

If the home is where dispositions form, the classroom is where they are reinforced or contradicted, and the question of how teachers and institutions respond to generative AI will determine which of those outcomes becomes the norm. The material on cognitive ownership, productive struggle, and contextual assessment covered in earlier chapters applies directly here, so this chapter will reference those arguments rather than revisit them. The focus is on what is specific to the institutional setting: the divide between how educators and students are experiencing AI, the practical redesign of assessment and instruction, and the governance frameworks that will determine whether individual good practice can become something more durable.

The conversation about AI in schools has too often been framed as a single conversation, as though students and educators were encountering the same phenomenon from a shared perspective. They are not. Research on faculty and student perceptions of generative AI reveals a consistent and consequential gap. Students tend to view AI tools pragmatically, as part of the infrastructure of academic life, something as routine as a search engine or a word processor. Faculty are more likely to foreground risk: concerns about plagiarism, about the erosion of disciplinary skills, about how to maintain the integrity of assessments they have used for years (Chong et al., 2026). Both positions are defensible, but neither is accurate.

The institutional reflex has largely been restrictive. Policies have focused on detection, on clarifying what is and is not permitted, and on updating honour codes to include AI-specific prohibitions. These are not pointless responses, but they address symptoms rather than the underlying condition. The transition that institutions need to make is from a policing orientation to a pedagogical one, and that transition requires more than updated conduct codes. It requires a different theory of what education is for in an AI-saturated world.

Redesigning Assessment for What AI Cannot Do

The core argument for assessment redesign has been developed in earlier chapters: when the final product of an assignment can be generated by a capable AI tool, the final product is no longer a reliable measure of the student's understanding. This section does not repeat that case but addresses the practical question of what assessment can look like when it is designed with this constraint explicitly in mind.

The most important conceptual move is the distinction between structural competence and semantic originality. Large language models are highly capable at the

former: they organize text according to disciplinary conventions, maintain consistent register, and produce work that looks like the work of someone who knows what they are doing. They are considerably weaker at the latter. AI-generated text tends toward the semantically generic, toward the patterns most frequently instantiated in the training distribution rather than toward the specific, argued, personal claim that reflects a particular mind working through a particular problem (Inoshita et al., 2026). An assessment that rewards structural competence will be easily replicated by AI. An assessment that rewards semantic originality, that asks a student to make a specific argument no one else in the cohort has made, is substantially more resistant.

One concrete implementation of this principle is the Argument Rarity-based Originality Assessment framework (AROA), which quantifies how rare a student's argument is relative to the full set of submissions in a cohort. In large undergraduate classes, the density of the reference corpus actually improves the reliability of this measure. A cohort of several hundred students submitting on the same topic provides a rich comparative set against which genuinely original arguments stand out from the generic patterns that AI characteristically produces (Inoshita et al., 2026). This does not replace human judgment; it provides a signal that can direct attention toward work warranting closer review. In smaller classes, the framework is less statistically robust and works better as formative feedback than as summative ranking.

Portfolio assessment offers a complementary approach grounded in process rather than originality scoring. When students submit initial outlines, multiple drafts, and reflective commentaries documenting how their thinking evolved, the trail of development is substantially harder to fabricate than any single submission, because the intellectual journey has to be coherent across multiple stages. Faculty reviewing portfolios are assessing not just what a student concluded, but how they got there and what they reconsidered along the way (Dau et al., 2026). Oral defences add a real-time verification layer that even the most sophisticated AI use cannot substitute: a student defending their argument in a live conversation with an examiner must demonstrate understanding in the moment, without recourse to any tool.

The combination of contextual complexity in task design, discussed in the previous chapter's treatment of the home environment, with process documentation and oral verification, creates an assessment ecology that is far more resistant to AI substitution than any individual measure alone. It also, not incidentally, produces better learning. Students who are required to document their thinking process, revise in response to feedback, and then explain their conclusions tend to understand the material better than those who produce a single polished submission, because the process itself does the pedagogical work (Chong et al., 2026).

Critical Thinking as a Teachable Practice

The case for teaching critical thinking explicitly has been made across several chapters of this book. What this institutional context adds is the question of what that teaching looks like when AI is simultaneously a subject of the curriculum and one of the tools available to support learning.

Research on short, structured critical AI engagement programmes finds measur-

able improvements in students' ability to verify information, detect inconsistency, and distinguish between AI-generated assertions and independently substantiated claims (Ngo Cong-Lem, 2025). These gains are not automatic consequences of AI exposure; they require explicit instruction, structured reflection, and the kind of iterative practice through which cognitive habits consolidate. A student who has been shown how to cross-reference an AI output against reliable sources and who has practised doing so across multiple tasks develops a different orientation to AI than one who has simply used it frequently.

AI literacy is the strongest predictor of the quality of critical thinking in AI-augmented environments. This is not literacy in the narrow technical sense of knowing how to operate a tool, but in the fuller sense of understanding how the tool generates its outputs, what its systematic weaknesses are, and where its confident-sounding text is most likely to be unreliable (Ramadhan and Susanti, 2026). No matter how advanced a model is, there's no clear consensus on a metric that can reliably be used to calibrate trueness within the internal state of AI, because of the emergent complexity of the data corpus used to train the models, the model's internal beliefs and dependency on semantics and context. Students who understand that large language models optimise for plausible-sounding outputs rather than for truth, and understand what hallucination means and roughly when it is more likely to occur, and who recognize that AI systems trained on existing text tend to reproduce the biases embedded in that text, are equipped to be genuinely critical users. Those who lack this knowledge are likely to accept fluent AI output as authoritative even when it is wrong (Mitrulescu et al., 2025).

The most effective pedagogical framing is one in which AI is positioned as a Socratic interlocutor rather than an answer engine. Tools that respond to student input with questions, that identify what a student has not addressed rather than supplying the missing content, and that require students to articulate their reasoning before receiving feedback, keep the cognitive work where it belongs (Lee et al., 2025). This is a design choice about how AI is deployed in instructional contexts, not a property of the tools themselves. The same underlying model can function as an answer machine or as a dialectical partner depending on how it is prompted and how its outputs are used.

Specific cognitive strategies matter alongside the broader AI literacy framing. Students benefit from being explicitly taught to treat AI output as a hypothesis to be tested rather than a conclusion to be accepted, to locate and interrogate the assumptions embedded in AI-generated text, and to compare AI outputs across multiple prompts rather than accepting any single response as definitive (Ngo Cong-Lem, 2025). These are teachable skills. They are not complex. But they require deliberate practice in conditions where the goal is evaluation rather than production, and they will not emerge from unguided AI use. **Making these strategies a habit is key.**

The Educator in Transition

Everything described above requires teachers who understand what they are asking students to do and why. That is not a given. Faculty who received their professional training before generative AI became widespread were not taught to think about AI

literacy, about assessment design as a response to AI capability, or about the specific ways that different prompt strategies produce different pedagogical outcomes. Providing them with a list of approved tools and an updated academic integrity policy does not fill that gap (Huang, 2024).

Effective teacher professional development for the AI era has several components that research identifies as distinct from general technology training. One is competence in responsible prompting, the ability to design prompts that produce pedagogically useful AI behaviour rather than simply fluent output, and to help students do the same (Zafar et al., 2025). Another is the ability to critique AI-generated content against specific curricular objectives, because AI outputs are characteristically generic and require significant adaptation to serve the precise learning outcomes a teacher has in mind. The teacher's role in an AI-integrated classroom is shifting from primary knowledge transmitter toward something more like *cognitive coach*: a figure who is present not primarily to deliver information, which AI can do at scale and in tailored form, but to create the conditions under which students encounter difficulty productively, provide the human judgment that contextualises AI feedback, and model the intellectual habits that the curriculum is supposed to develop (Chadha, 2024). This is not a diminished role. It is a more demanding one, which changes the profile of a teacher into a higher, more inclusive domain. And it requires institutional support rather than just individual adaptation, with incentives that should begin with salaries to justify such broader pedagogical scope.

At the policy level, institutions face a structural challenge. Most governance frameworks for AI in education were developed reactively, in response to the immediate concern about academic dishonesty, rather than proactively as part of a considered approach to pedagogical change. Research on governance in higher education contexts identifies a recurring pattern: institutions announce policies on AI use before they have developed the professional development infrastructure to support those policies, and before they have thought through the equity implications of differential access to capable tools (Thaldar et al., 2025). The result is a policy vacuum in which faculty are expected to enforce rules they were not consulted on, using detection tools whose limitations and effects in students may not be fully understood, in the context of institutional frameworks that have not been updated to reflect how AI is actually being used by students.

Closing this gap requires moving governance from a reactive to a proactive mode. Institutional oversight committees with genuine faculty representation, professional development programmes that are ongoing rather than one-off, and equity-focused access policies that address the freemium problem discussed in the previous chapter are the structural levers available (Owidi et al., 2025). And although history doesn't repeat itself it often rhymes. This reactive vs. proactive stand has been a historical tug-of-war between education legislation and its teachers, where the latter argue that adding "monkey patches" (borrowing from computer science jargon referring to the action of adding, modifying or suppressing the default behavior of existing system without changing the original code) leads to undesirable learning outcomes for their students. Multiple communications and words of advice from teachers in the Vancouver School Board have pointed at this complex social endeavor, where change is never simple to implement and none of it happens automatically or without struggle. But

the alternative, in which the classroom continues to transition (with AI as a catalyst for this change) under pressure without institutional support, is already producing very noticeable cracks in a previously fragmented, inconsistent and inequitable landscape.

To end our section with a positive note, I should also mention some of the proactive steps in legislation going on in Canada in 2026 that seeks to address this problem. Canada's federal "AI for All" strategy, launched in June 2026, commits \$30 million through CanCode to deliver AI literacy training from kindergarten to Grade 12, alongside a National AI Literacy Initiative targeting a million post-secondary students and classroom learning kits for educators ([Government of Canada, 2026](#)). In British Columbia, the Ministry of Education has published guidance affirming that teachers retain professional judgment over AI tool use and that school districts must maintain ownership of all student data generated through those tools ([BC Ministry of Education and Child Care, 2024](#)). The Vancouver School Board followed days after the federal strategy launch with the rollout of supervised AI accounts for students aged 13 and up ([Dickson, 2026](#)). At the provincial level, the Ontario Teachers' Federation published a substantive discussion paper calling for GenAI policy to be developed in genuine collaboration with teachers, students, and unions rather than handed down from above ([Ontario Teachers' Federation, 2026](#)). These are very proactive steps with mixed opinions, but at the very least shows an increase acknowledgment for change. The Canadian Teachers' Federation has welcomed the direction while noting the scale remains insufficient: Canada currently ranks 44th of 47 countries on AI literacy, and the strategy as written would train roughly 3,000 of the country's 420,000 public educators ([Canadian Teachers' Federation, 2026](#)). The framework is being built, but whether it is built quickly enough, and with enough teacher involvement and support remains an open question.

Preparing Educators: Before and After the Classroom

Research on teacher education and AI integration identifies a more precise target than general digital literacy: AI-TPACK, the intersection of a teacher's knowledge of AI tools and their ability to deploy those tools purposefully within a specific subject domain to achieve defined learning outcomes (Rothinam et al., 2026). General AI familiarity does not automatically transfer to effective AI pedagogy. A teacher who knows how to use a generative AI tool for their own writing does not necessarily know how to design a history assignment that uses AI productively without bypassing the historical reasoning the assignment was meant to develop. Making that design judgment requires a distinct competency, one that sits at the intersection of technical knowledge, curricular understanding, and pedagogical craft (Said et al., 2023).

Practical frameworks for building these competencies have begun to emerge. The D-COMP model, developed for pre-service digital competence training, emphasizes a shift from teaching about technology to teaching through it, connecting digital capability with critical pedagogical practice rather than treating technical training as its own endpoint (Ahmad et al., 2025). Implementation is structured in phases: a foundation phase focused on curriculum mapping and institutional readiness assessment; an implementation phase involving project-based application of AI tools in simulated and real classroom environments; and a consolidation phase in which competence is evaluated through rubrics and micro-credentials and policy is revised accordingly. What makes this structure effective is that it is iterative rather than one-off. The speed at which AI systems are developing means that any training delivered once and not revisited will become outdated within a year. Professional development for AI is an ongoing commitment, not a certification (Manju et al., 2026).

In-service development faces an additional challenge: the gap between educators who can integrate AI fluidly and those who struggle to align AI-generated content with specific curricular objectives is already wide, and it is not closing on its own (Huang, 2024). Bridging this gap requires sustained institutional investment, not the expectation that individual teachers will update themselves on their own time. In science education specifically, development must move beyond technical enhancement toward what researchers are calling *epistemic responsibility*: training teachers to evaluate AI outputs across both factual and ethical dimensions, so that they can model this evaluation for students rather than simply endorsing whatever the tool produces (Mohammadipour, 2025).

The most productive professional development, at both pre-service and in-service levels, is grounded in co-design. Curricula developed by researchers without teachers tend to be disconnected from actual classroom dynamics; curricula developed by teachers without researchers lack the theoretical grounding that makes them transferable. Cross-disciplinary co-design processes that bring both together have produced genuinely useful AI curricula, but they naturally also surface genuine tensions, where one side prioritizes technical accuracy (i.e. an operative) vs. ethical and rhetorical (theoretical) dimensions of what AI is saying (Xie et al., 2024b). These tensions are not obstacles to resolve but rather the very reason co-design is valuable, because they produce curricula that neither group would have arrived at independently, generating a **hybrid model**.

Restructuring Active Learning for the AI Age

Project-Based Learning (PBL) has long been favoured in educational research because it places students in contact with real problems and requires them to apply knowledge rather than merely demonstrate recall. Its vulnerability in the AI era is that many PBL tasks are structured around producing professional-looking outputs, reports, plans, proposals, and AI is now highly capable of producing exactly those things. A project culminating in a deliverable that an AI could generate in four minutes is not providing the learning experience it was designed to provide.

The structural fix, as noted in the discussion of assessment redesign, is to embed contextual complexity that is genuinely resistant to AI substitution: experiential dimensions drawing on the student's own observations, local or real-time data the AI cannot access, and ethical judgments that require the student's own values rather than a generic summary of ethical frameworks. A project asking students to analyse urban planning in their own neighbourhood, interview residents, and argue for a specific decision on the basis of that evidence could be drafted, in support with the teacher, using prompts and followed by a hands-on experience based on its deployment, ensuring the *cognitive ownership* while reducing the *preparatory burden* we discussed in Chapter 2. When combined with other strategies like AROA the instructor gets a tool to evaluate PBL based on evidence AI cannot replicate.

A very recent development in the current research that I think is worth mentioning points at the articulation of ***design-phase logic***: a useful model of where AI should be integrated into a project and where it should be deliberately excluded. The SuperSkillsStack framework, developed in design education, identifies four capacities that must remain with the student throughout a project: Agency, Domain Knowledge, Imagination, and Taste (Huang and Poon, 2026). AI can legitimately accelerate the early stages of a project, the brainstorming, information synthesis, and problem-framing phases where speed is valuable and the stakes of any individual idea are low (i.e. the dossier). But as a project moves toward implementation, each of these four capacities becomes load-bearing. **Domain knowledge** is needed to evaluate whether an AI-generated suggestion actually works in the specific context. **Taste**, defined as evaluative judgment, is needed to distinguish meaningful solutions from plausible-sounding but superficial ones. **Agency** is needed to decide when AI should be consulted and when the thinking must be done without it. **Imagination** is needed to generate the genuinely original contribution. These are not skills to be developed after students have learned to use AI tools. They are the skills that must be in place before AI use becomes productive rather than substitutive.

A concrete example of this logic in practice is the Enriched Twin programme, which combines drone-based data capture, 3D modelling, and agentic AI in a digital heritage project (Cui et al., 2026). AI is not the product of the project but one of its instruments. Students use it within a multi-stage workflow that also requires physical fieldwork, direct engagement with cultural objects, and ethical reasoning about how heritage is represented and to whom. When AI is embedded as one step in a broader experiential sequence rather than positioned as the answer generator, it functions as a genuine scaffold rather than a bypass.

Inquiry-Based Learning (IBL) in contrast to PBL, faces the same structural chal-

lenge from a different angle. IBL depends on students experiencing genuine uncertainty, forming hypotheses, testing them, and revising. The automation paradox documented in the SmartFlash research we discussed in Chapter 3 applies directly: when AI absorbs the initial synthesis and organisation work that inquiry requires, students lose the productive confusion that drives the formation of good questions. The Socratic Mind platform addresses this by design. Rather than providing answers, it identifies where a student's understanding is incomplete and asks questions that point toward the gap (Lee et al., 2025). The platform is not trying to make inquiry easier; it is trying to make the difficulty productive rather than simply discouraging.

Human-AI Collaboration as a Principled Practice

The phrase human-AI collaboration is used in different context, even in educational writing. We define it as a handful of design principles, focusing on a *successful* interaction (i.e. boosting critical thinking) an individual experiences by integrating these technologies into their learning and task-completion workflows.

The first principle is that effective collaboration requires **domain knowledge** on the human side before AI is introduced. Research on AI use consistently finds that experts in a theoretical framework use AI to accelerate and extend work they already understand, while unexperienced users use it to avoid developing the understanding they lack (Khan, 2025). The practical implication is that AI should generally not be the first resource a student consults for an unfamiliar problem (in an increasingly AI-dependent browser interaction, perhaps this means removing the computer altogether). The attempt at independent engagement, even a flawed and incomplete one, establishes the cognitive state that makes AI interaction genuinely useful.

The second principle is **productive friction**. Where the first principle concerns timing (independent engagement before AI), this one concerns design: assignments should build in moments where students must demonstrate understanding that AI cannot provide on their behalf. Requiring students to apply an AI-generated text to a case the AI has not been given, or to find the flaw in a deliberately erroneous AI response, or to extend an AI-produced argument into territory it has not addressed, all create moments of genuine cognitive engagement within an AI-assisted workflow (comparable to talking with a colleague or tutor). These are not adversarial gestures toward AI. They are design choices about where in the workflow human thinking must happen.

The third principle concerns modification via **cognitive ownership**. As established in the discussion of the SmartFlash findings, the act of editing AI-generated content is not merely about error correction. It is the mechanism by which external information is transformed into personal knowledge. The non-negotiable element here is justification: the student must be able to articulate why they changed what they changed, which requires having formed an independent view that the AI output was being measured against.

The fourth principle, and the one that encompasses the others, is what the research literature calls **Human-in-the-Loop** (HL) as a design pillar rather than a workaround. HL is sometimes framed as a corrective mechanism: humans check AI

output when something seems wrong. In educational design, it needs to be structural: human agency is built into every stage of the workflow, not available as an intervention when things go wrong (Dranias and Whitley, 2026). This means assessing process rather than only product, using tools like provenance logs (which themselves can be automated) or annotated drafts where students document their interactions with AI and their reasoning about the outputs, and designing tasks where the human contribution is the part being evaluated rather than the part that happens to be left over after AI has done what it can do.

CHAPTER 6

Choosing a Path

The preceding chapters have examined how AI affects the development of thinking capacity and what the home and classroom can do to protect it. This chapter asks a different question, one that follows naturally from everything before it: given what AI is doing to knowledge work, which educational and career paths are genuinely worth investing in? This is practical territory. Parents and students making decisions about degrees, majors, and professional development cannot wait for the research to fully settle; they need frameworks that work with the uncertainty rather than against it. What follows is an attempt to provide exactly that, grounded in what labour market research, cognitive science, and educational economics currently show.

What AI Is Actually Doing to Knowledge Work

The public discussion of AI and employment has oscillated between two equally unhelpful positions: the catastrophist view, in which automation will eliminate most jobs within a decade, and the dismissive view, in which AI is simply another productivity tool, no different in kind from the calculator or the word processor. The research sits in more productive territory.

What the evidence shows is that AI is reshaping work through a process researchers call task recomposition. Jobs are not simply deleted; they are broken down into constituent tasks, and those tasks are treated differently depending on their nature. Tasks that are procedural, standardisable, and follow predictable logical patterns are automated or substantially assisted by AI. Tasks that require high-level judgment, novel synthesis, contextual interpretation, or genuine human relationship are either augmented by AI or become more economically valuable precisely because they resist automation (Chiriță et al., 2025). This distinction between routine and non-routine cognitive work is the most important single concept for thinking about career durability in the current moment.

The sectoral picture is uneven. Financial and business services, specific functions within legal services, translation and interpretation, and data processing roles are among those facing the highest near-term substitution pressure, because their core functions involve the kind of language processing and information retrieval at which large language models excel (Cai, 2026). Customer-facing administrative roles and mid-tier coding tasks follow a similar pattern. Healthcare and education, by contrast, are more typically associated with augmentation rather than substitution: AI assists professionals in these fields but does not replace the human judgment and relationship

work that defines them (Chiriță et al., 2025). The important caveat is that even in augmentative sectors, specific tasks within roles are increasingly automated, which means the professional who understands which parts of their work AI is now handling, and can focus their own effort accordingly, is in a different position from one who has not thought about it (Chen, 2025).

The macroeconomic picture is less alarming than the headlines suggest. Panel data analysis across developed economies finds a statistically significant association between AI patent growth and modest decreases in overall unemployment rates, perhaps consistent with the view that productivity-enhancing technology tends to create demand for labour even as it displaces specific roles (Yildiz et al., 2025). New positions in AI governance, explainability, ethics review, and human-AI interface design are emerging alongside the automation (Ejjami, 2024). What the research makes clear, though, is that these new roles are not automatic replacements for displaced ones; they require different competencies, and they disproportionately reward people who have built genuine intellectual capital rather than those who have simply operated tools. Doing it otherwise has already shown massive losses in capital, especially in companies that took the risk to replace intellectual workers with automation.

One finding from the emerging economics of AI and data deserves special attention: the work of verifying, correcting, and updating AI systems is not a temporary phase in the transition to full automation. Because the world changes continuously, AI systems face persistent drift as edge cases accumulate, standards evolve, and context shifts. The human labour required to catch these failures and keep AI systems aligned with real-world conditions is a structural feature of the AI economy, not a gap to be eventually closed (Ansari et al., 2026). This is practically significant. It means that domain expertise, the capacity to recognise when an AI output does not fit the actual situation, will remain economically valuable even in highly automated environments. The expert is not simply the last person to be automated; they are the person the automated system requires in order to function reliably. The specialized worker therefore has more resources to work at higher levels within the semantics of their company (i.e. building systems that interlock, where tasks down the road are semi or fully automated).

The Taxonomy of Durable Skills

If the economic argument above is correct, the question becomes specific: what skills remain durable as AI handles an expanding share of routine cognitive work? The research points to a layered answer.

At the foundation are what some researchers term invariant skills: capabilities that remain relevant across technological shifts because they concern the quality of human reasoning rather than the execution of specific procedures. Strong research and analytical capability, the ability to synthesise complex and contradictory information into coherent argument, and the capacity to apply logical reasoning to problems that have no standard procedural solution are the clearest examples (Chiriță et al., 2025). These are not particularly glamorous skills because they rely on pre-organization and methodology, which may seem “boring” in the world of business, especially when seen through the eyes of the public and media. They are, however, the ones most consis-

tently identified across different disciplinary traditions as the core of educated human judgment, and they are the ones AI handles least well.

Above this foundation sit higher-order competencies that depend on lived experience and genuine situational understanding. Ethical judgment in ambiguous contexts, the ability to read a social situation and respond with appropriate nuance, and the kind of original synthesis that shifts how a field understands a problem are capabilities AI can simulate in limited ways but cannot authentically exercise, at least in the foreseeable future. A language model can produce text that resembles ethical reasoning; it cannot bear the weight of a genuine moral decision in a real professional context (Ngo Cong-Lem, 2025). The economic value of these competencies is increasing because AI makes the routine cognitive work they used to accompany cheaper, which makes the specifically human contribution more visible and more valuable.

The most important practical concept here is the difference between usage familiarity and critical familiarity (Singh et al., 2026). *Usage familiarity* means knowing how to operate an AI tool effectively: constructing good prompts, recognising the general capabilities of different systems, integrating AI output into a workflow. This is the sort of information where YouTube videos are a great platform to learn. *Critical familiarity* sits at a higher level of reasoning, and means understanding what the tool does not know, where its outputs are most likely to be unreliable, and how to catch errors that would not be caught by someone who simply trusted the output. The research on AI use in professional and educational settings consistently finds that the confidence-competence gap is largest among frequent users who have developed high usage familiarity but low critical familiarity (Koch, 2026b). The durable professional is not simply the one who uses AI most, but the one who uses it while maintaining the independent judgment to know when it is wrong or not the right tool for the task.

A personal example could help illustrate the difference more concretely. In early 2026, there was enormous excitement in the software development world about agents: programs that could reason about a task, make small decisions, and hand off results to the next step in a pipeline. Some developers (including me) were inserting agents into nearly every workflow, often simply because they could. At the time, I was working on a program that generated virtual three-dimensional landscapes, producing terrains that looked like hills, caves, and valleys from a set of tunable settings: how smooth or rough the surface was, how many cave systems it contained, how the ridges and valleys were distributed across the terrain. I was curious whether agents could do something useful here, so I set them up to reason about those settings and propose combinations that would generate landscapes worth keeping. The program ran. The agents made decisions. But the results were computationally expensive to produce, and after all that effort, the landscapes looked strikingly similar to one another, maybe even boring. They lacked the variety I was hoping for. I sat with this for a while, attempting to “convince” the agents to do what I needed, and in the process of doing so I realized that instead of agents I could simply randomise each setting independently, generate thousands of landscapes in a fraction of the time, look through a couple of them myself, and keep the ones I found interesting. No reasoning required. No agents needed. The human eye, applied to a large random sample, was doing the real selection work more cheaply and more effectively than the elaborate automated alternative and it was exactly what I wanted. The lesson was not that agents are useless. It was that I had

reached for a sophisticated tool before asking whether the problem actually required it. That question, “is this tool the right one for this task, and do I understand it well enough to know when it is failing me?” is the heart of critical familiarity.

The SuperSkillsStack described in the previous chapter, combining Agency, Domain Knowledge, Imagination, and Taste, provides a useful shorthand for this highest layer of durable human capital (Huang and Poon, 2026). The four elements are worth restating briefly in a career context rather than a pedagogical one. Agency is the capacity to decide when AI should and should not be part of a workflow. Domain Knowledge is the grounded expertise that makes it possible to evaluate AI outputs against reality. Imagination is the ability to generate ideas that do not exist in the training data of any current model. Taste is the evaluative judgment to know which ideas are meaningful in a specific professional and cultural context. Taken together, these are not skills that AI can develop on a person’s behalf. They require deliberate practice, genuine intellectual investment, and the kind of productive friction that earlier chapters argued must be preserved in educational settings, and now get translated into tangible consequences within professional profiles and industries.

The Specialisation–Generalism Paradox

Career advice has traditionally organised itself around a binary choice: become a deep specialist, or develop broad cross-domain capability. AI is making this binary increasingly obsolete, but not in the way that is sometimes suggested.

The vulnerability of narrow specialisation is real but specific. The risk is not that domain expertise becomes worthless; it is that the specialist who calibrates their professional confidence against AI-assisted outputs gradually loses the independent judgment that genuine expertise requires (Khan, 2025). A specialist in translation, financial analysis, or code review whose skills have atrophied through over-reliance on AI tools faces not just a performance decline but a reduced capacity to catch the errors that matter most, the ones that require domain knowledge to recognise as errors at all. This is the crutch dynamic described in Chapter 2, operating at professional scale over years rather than at student scale over a semester.

The transformation of translation is instructive. Basic translation of routine documents is now within the reach of capable AI systems. What remains genuinely valuable in translation work is precisely the human touch: cultural interpretation, the detection of implication and tone that depends on situational context, ethical judgment about how to handle sensitive or contested material, and the kind of nuanced client communication that determines whether a translation serves its actual purpose (Shi et al., 2025). The translator who has maintained and developed these capabilities alongside AI tools is more productive than ever; the one whose capabilities have narrowed to operating translation software is in a genuinely precarious position. Legal work follows a similar pattern. AI can identify relevant precedents, summarise arguments, and flag inconsistencies with high efficiency, but the judgment about how to apply a principle to a genuinely novel situation, and the professional responsibility that attaches to that judgment, remains with the human practitioner, including the consequences of such decisions (Isakov et al., 2024).

The strategic response is neither pure specialisation nor broad generalism but what has been called the T-shaped profile: deep expertise in one domain combined with enough cross-domain literacy to collaborate effectively across fields and to recognise when tools and insights from adjacent areas apply to one's own. In practice, this means developing orchestration skills alongside domain expertise (Bhati, 2026): the ability to decompose complex goals into components, to guide AI effectively toward specific outcomes, to review AI output critically, and to make sound judgments about when to delegate and when to retain cognitive ownership. These skills are not natural by-products of AI use; they require explicit cultivation. The newcomer who uses AI tools without developing the orchestration capacity that makes those tools useful in demanding professional contexts is, as the research on two-track skill markets documents, at a persistent disadvantage relative to the experienced practitioner who has both (Khan, 2025).

The Return on Investment of a Degree

The most practically loaded question this book's audience faces is whether a four-year university credential remains a sound investment in an economy where AI is automating knowledge work. The most honest answer is perhaps: it depends what the degree is actually building.

The traditional case for a university degree rested on two arguments: skills acquisition, meaning the graduate actually knows and can do things they could not before, and signalling, meaning the credential communicates cognitive ability and discipline to prospective employers. AI is weakening the second of these without necessarily affecting the first, and the weakening is specific. When students use AI to bypass the intellectual work that a degree is supposed to require, and the resulting credential reflects the AI's capabilities rather than the student's, the signalling value of the credential degrades for everyone who holds it (Chong et al., 2026). This is not a hypothetical; it is the pressure that assessment redesign, discussed in the previous chapter, is trying to address.

The more substantive question concerns which disciplines are building durable capabilities under the new conditions. The conventional wisdom that technical degrees guarantee economic security and humanities degrees do not is increasingly inadequate as a guide. AI is automating routine coding, data entry, basic financial analysis, and elements of technical report writing, which means that a STEM credential whose value rests primarily on these capabilities is less durable than it appeared a decade ago (Cai, 2026). Technical education that emphasises system design, Human-in-the-loop governance, and the orchestration of complex AI workflows is more durable, because these are the skills the technical economy now requires and cannot readily automate (Dranias and Whitley, 2026).

Liberal arts, social sciences, and humanities face their own version of this revaluation. The argument against these fields has usually been that they do not produce immediately employable graduates. The argument for them has been that they build the invariant skills most resistant to automation: complex argument, ethical reasoning, cultural interpretation, and synthesis across conflicting sources. That second argument

is stronger now than it was when it was first made, because the premium on exactly these capabilities is rising as AI handles more of the routine cognitive work they used to accompany (Chiriță et al., 2025). The most durable educational investment, regardless of discipline, is one that explicitly cultivates higher-order reasoning and critical familiarity with AI, not as additions to a curriculum but as its central purpose.

The concept of an augmentation premium captures something important here. The research finding that skilled workers who use AI for augmentation tend to see increased productivity and income, while lower-skilled workers who interact with AI as a substitute tend to face displacement, implies that the return on a degree increasingly depends on whether the graduate is capable of the former rather than the latter (Zhu, 2025). Building that capability requires the metacognitive awareness and critical habits that earlier chapters argued must be preserved in educational settings. A degree that produces graduates who know how to use AI tools but not how to evaluate them critically, and not how to maintain the independent judgment that augmentation requires, is producing graduates who are vulnerable to the very displacement the degree was supposed to protect against.

Strategic Adaptability and the Career as Iterative Project

There is a structural feature of AI-integrated professional environments that changes how career development should be planned, and it is worth stating plainly. The half-life of specific technical skills is shortening. AI systems are updated, deprecated, and replaced on timescales that do not align with the traditional model of front-loaded education followed by decades of application. A professional who invests in mastery of specific tools without maintaining the foundational capabilities that make new tools learnable is facing a trajectory of progressive skill obsolescence (Ait Soudane and Moutahid, 2021). This is not a new problem, but AI is accelerating it substantially.

The practical implication is that the career is better understood as an iterative project than as a linear progression toward a stable destination. Strategic pivots, periodic reinvestment in new capabilities, and the willingness to move toward roles that AI cannot yet replicate are structural features of durable professional life in the current environment (Xu, 2024). This does not mean permanent instability or the abandonment of deep expertise. It means understanding that expertise in the application of a specific tool is not the same as expertise in a domain, and that the latter transfers across technological generations while the former does not.

The human-AI flywheel captures the productive version of this dynamic. When humans use AI effectively for augmentation, they produce higher-quality work, which generates better training signal for AI systems, which enables more ambitious augmentation, creating a cycle of mutual reinforcement in which human capability and AI capability develop in tandem rather than in competition (Farach et al., 2025). This is a massive, global-scale Human-in-the-loop integration. Participating in this flywheel rather than being displaced by it requires staying on the augmentation side of the augmentation-substitution boundary, and staying there requires maintaining the independent judgment and critical familiarity that make genuine oversight possible.

A Decision Framework for Students and Parents

Everything in this chapter converges on a practical question: how should students and parents think about educational investment right now? The following framework does not pretend to eliminate uncertainty, but it identifies the considerations that matter most.

The first criterion is whether a curriculum builds domain expertise before, or alongside, AI tool proficiency. The research on expert versus novice AI use is consistent: the professional capable of evaluating AI output has domain knowledge that predates AI assistance (Khan, 2025). Programmes that introduce AI tools before the foundational domain knowledge is in place tend to produce graduates with high usage familiarity and low critical familiarity, which is precisely the profile most vulnerable to the confidence-competence gap.

The second criterion is whether the curriculum cultivates the invariant skills, research capability, complex synthesis, and deep reasoning, not as complements to the main programme but as central outcomes. This is less about discipline than about pedagogical design. A technical programme that builds these skills through its subject matter is more durable than one that does not, and a humanities programme that fails to build them provides poor return of investment (ROI) regardless of the abstract case for the discipline.

The third criterion is whether the programme prepares students for lifelong adaptation. The specific skills that will be most valuable five years after graduation are difficult to predict; the capacity to acquire new skills, to recognise when current capabilities are becoming obsolete, and to pivot toward emerging needs is itself a durable asset (Ejjami, 2024). Programmes that teach students how to learn, and that cultivate metacognitive awareness of one's own knowledge and its limits, are providing something that will compound over a career in ways that specific technical training cannot.

For parents specifically, the most valuable guidance is to resist the pressure to optimise for current employment statistics. The fields with the highest graduate employment rates today are not necessarily those that will remain most valuable as AI continues to develop. The fields that develop the capabilities described in this chapter are the better bet, even when those bets look less immediately legible in salary surveys. A child who graduates with strong analytical reasoning, genuine domain expertise in something, and the critical familiarity to use AI without being replaced by it is better positioned for the coming decade.

The Liberal Arts and STEM: Why the Old Binary No Longer Holds

The perceived competition between technical and humanistic education has always rested on a mistaken premise: that technical skills produce immediately useful graduates while liberal arts produce intellectually interesting but economically risky ones. AI is dismantling this premise, but not by making technical skills less valuable. It is doing so by revealing that the skills long associated with humanistic training — careful reading, contextual judgment, ethical reasoning, sustained argument — are increasingly the differentiators within technical fields, not just alternatives to them.

The vulnerability introduced by narrow technical training was the focus of our previous section. What deserves more attention here is the positive case for hybridity, best illustrated by one concrete example.

Scott Eaton is a sculptor who spent years in rigorous classical figurative training, studying anatomy in the tradition inherited from the European ateliers — proportion, musculature, the underlying structure that makes a human form read as anatomically credible rather than superficially resembling one. When generative AI tools for image synthesis became capable, Eaton's response was not to resist them or defer to them. His classical background gave him something most users of those tools did not have: a precise internal standard for evaluating what the AI produced. He knew, from years of drawing and sculpting, when an AI-generated material was anatomically incorrect, and most importantly, on an idea of how to use it in his favour. He could direct the tools toward outputs that were anatomically meaningful in the semantics of the scene rather than simply biologically correct. The result was work that used AI's generative capacity while remaining genuinely his own. Only someone with his specific training could have produced it, regardless of what tools were available. Someone without that background using the same software would produce something generically competent, not something that bore the marks of deep expertise directed through technology.

This example is useful precisely because it is not about coding or data science. It is about what happens when any kind of genuine domain knowledge meets AI capability: the person with the knowledge can direct the tool toward meaningful results and evaluate its outputs against a real internal standard. The person without it is operating in a register the tool cannot inform them about. The “automated versus augmented” distinction (covered in the previous section as the central finding of the Automated but Atrophied research) is not primarily a distinction about which tools a person uses. It is a distinction about whether the person has the domain knowledge required to know what the tool should be producing (Khan, 2025). Eaton's case demonstrates this in a domain far from computer science, which is exactly why it is illuminating.

The practical implication for students choosing between technical and humanistic programmes is not that they should choose one over the other, but that they should choose the one in which they can build genuine depth, and then develop the critical AI literacy (the ability to evaluate, interrogate, and when necessary dismiss AI output) that makes that depth usable in an AI-augmented environment. The **Augmented Human Capital framework**, emerging from recent work in labour economics, makes the argument formally: the traditional model for estimating education returns underestimates the value of cognitive capabilities that are complementary to

AI, and overestimates the value of those that are substitutable by it (Espinal Maya, 2026). In plain terms: the return on depth that makes AI useful to you is increasing, and the return on surface competence that AI can replicate is declining.

Where STEM Trajectories Are Actually Heading

For students in technical fields, understanding the direction of change within their disciplines is more actionable than general career advice. Three trajectories illustrate the broader pattern.

Software engineering is undergoing the most visible transformation. AI coding assistants have already demonstrated significant productivity gains for developers, with less experienced programmers often benefiting most from the immediate augmentation (Pinto et al., 2023). But the direction of travel is not simply “AI writes code, developers review it.” The professional trajectory is moving toward system design, agent orchestration, and governance. By 2030, AI is projected to handle not just routine code generation but advanced automated bug identification and resolution through deep code analysis (Qiu et al., 2024). Autonomous testing agents are already capable of formulating and executing testing plans through conversation with developers in natural language, substantially reducing the manual effort of software verification (Feldt et al., 2023). What this means for someone choosing a software engineering programme now is that the curriculum’s value lies not in teaching specific languages or frameworks, which AI can generate competently, but in building the system-level thinking and critical judgment that makes it possible to direct AI toward correct, secure, and maintainable outcomes. A student who graduates able to decompose complex problems, specify what correct behaviour looks like, and evaluate AI-generated code against those specifications has what the field will require. A student who can write competent code but cannot articulate what makes it correct is developing a skill AI is acquiring faster than the degree takes to complete.

Test automation offers a sharper version of the same lesson. The professional role is already shifting from what researchers describe as “test creators” to “AI trainers and strategists”: professionals who define what quality means for a system, design test scenarios that probe for the right failure modes, and evaluate whether AI-generated tests are actually covering the important cases (Sai et al., 2025). This is not a less skilled role. It is a more conceptually demanding one, because it requires understanding what matters about a system, not just how to verify that it runs.

Data engineering is following a similar arc. Generative AI is transforming data infrastructure management, extending well beyond automation of data pipelines into how organisations approach the entire design of their data systems (Kurva et al., 2025). The growing skill requirement in the field is not data wrangling but governance: implementing validation frameworks, monitoring data quality and processing integrity, and ensuring that AI-generated solutions remain accurate as underlying data and business requirements evolve. This is work that requires understanding what the data should mean, not just how to move it.

Beyond software, the more interesting convergences are interdisciplinary ones. Bioinformatics is the clearest example of a field where the computing and biology boundary

has effectively dissolved. Modern biological research requires proficiency in sequence analysis, structure prediction, and computational tools that were, a generation ago, treated as specialist skills separate from the laboratory work. They are now integral to the laboratory work (Niepielko et al., 2021). A biology undergraduate who has not developed computational literacy is not simply missing a supplementary skill; they are missing a core component of how the field generates and evaluates evidence. At the research frontier, the development of AI architectures that draw on biomimetic principles — seeking solutions evolved by biological nervous systems to constraints that engineering faces in constrained environments like mobile devices or edge computing — is creating an interdisciplinary demand for researchers who can think across neuroscience, biology, and computer systems simultaneously (others, 2018). These roles do not fit the standard category of either biologist or engineer, which is precisely what makes them durable.

The common thread across these trajectories is the shift from execution to governance: from doing specific technical tasks to defining what the tasks should accomplish, evaluating whether AI is accomplishing them, and intervening when it is not. The electronics curriculum redesigns currently being piloted align with exactly this shift, moving from component-level instruction toward integration of AI, automation, and systems thinking (Tomun et al., 2025b). The student who leaves a technical programme with only execution skills, and without the conceptual framework to govern AI systems, is entering a labour market that is moving away from them.

The University's Own Transition

Universities face a version of the same challenge they are asking students to navigate. The traditional university business model — concentrated knowledge delivery in a four-year period, followed by a credential that signals readiness for professional life — is under the same automation pressure as the professional roles it has historically prepared people for. The response, predictably, has been uneven: some institutions have updated assessment frameworks and invested in AI literacy, while others have focused primarily on detection and restriction (Pokotylo, 2025). The latter approach is the academic equivalent of trying to remain competitive by banning the tools your competitors are using.

The more promising institutional direction treats AI integration as a pedagogical opportunity and an operational one simultaneously. On the operational side, AI can automate administrative processes, enable predictive analytics for student support, and personalise services in ways that free faculty from tasks that do not require their expertise (Malik et al., 2025). The goal is not to reduce faculty roles but to redirect faculty time toward what only faculty can provide: direct intellectual engagement, mentorship, and the kind of contextual judgment that AI cannot supply.

The more substantive shift is pedagogical. The Augmented Human Capital framework argues that the value of a degree is increasingly determined by how well it develops capabilities that are complementary to AI rather than substitutable by it (Espinal Maya, 2026). For universities, this implies moving assessment away from content reproduction — which AI handles — toward process evidence, contextual application,

and demonstrated reasoning capacity, exactly the redesign discussed in Chapter 5. It also implies structural partnerships with industry and practitioners to ensure that curricula track the actual direction of skill demand rather than lagging it by several years (Tomun et al., 2025a).

The equity dimension is non-negotiable. When access to premium AI models is determined by ability to pay, the institutions that can afford enterprise licences are providing a better educational environment by definition (Singh et al., 2026). Universal provision of capable AI access within institutions, treated as infrastructure equivalent to library access, is the structural commitment that prevents the university itself from becoming a reproduction mechanism for existing advantage.

The Red Queen and the Career as Continuous Adaptation

In evolutionary biology, the Red Queen hypothesis holds that species must keep evolving continuously just to maintain their current fitness, because the organisms they depend on and compete with are also evolving. Standing still, in an evolving environment, is the same as falling behind. Lewis Carroll had the Red Queen tell Alice that it takes all the running you can do to stay in the same place.

The analogy to professional development in the AI era is uncomfortably precise. The floor of what AI can do is not static. The cognitive tasks that were safely human two years ago have, in a number of cases, moved into AI's competence range. The response to this is not to run harder at the same tasks, but to develop the higher-order capabilities that the current state of AI cannot reach, with the understanding that the horizon will keep moving. This is not a counsel of despair. It is a description of what learning has always been for, stated more urgently than usual.

Skill obsolescence is already a documented source of professional insecurity, and it predates AI (Ait Soudane and Moujtahid, 2021). What AI changes is the rate. The half-life of specific technical knowledge is shortening as model capabilities advance, new tools appear, and the tasks that were hard become routine. A professional who treats their education as a completed asset rather than a living capability is in an increasingly vulnerable position, not because their knowledge becomes wrong, but because the economy's interest in it diminishes faster than it used to (Xu, 2024).

The productive response is a career structured as a series of deliberate capability investments rather than a linear progression toward a stable destination. At the undergraduate level, the priority is building the foundational thinking capacity and domain depth that make future learning possible. At the postgraduate and early professional level, the focus shifts toward bridging theoretical foundation and applied expertise, using AI-enhanced simulation and immersive practice to compress the distance between learning and professional reality (Yashchuk et al., 2025). At the mid-career level, the challenge is maintaining the self-assessment capacity to recognise when capabilities are becoming obsolete, and the institutional access to reskilling that makes adaptation possible rather than merely theoretically desirable (Mahmud et al., 2024).

The human-AI flywheel described in the previous section provides the optimistic framing of this dynamic: humans who engage seriously with AI tools produce better

outcomes, learn from the interaction, and develop more sophisticated ways of directing the tools, which enables more ambitious collaboration, which develops their capabilities further. The prerequisite for entering this flywheel, rather than being displaced by the tools, is the domain expertise and critical judgment that make meaningful AI direction possible in the first place. This is the Red Queen in productive form: the professionals who are continuously developing their judgment and expanding their domain understanding are not just maintaining their position. They are compounding.

In this framing, **the degree is not the destination**. It is the first serious investment in a capability that requires continuous reinvestment. The universities, employers, and individuals who understand this are building something qualitatively different from those who treat the credential as a terminus. For students and families making decisions right now, the most important question about any educational programme is not “what job does this lead to?” but “does this programme develop the kind of thinking that will allow me to keep learning well?” The Red Queen does not reward the fast. She rewards the adaptable.

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